

FANUC Robot **series**

R-30iB/R-30iB Plus CONTROLLER

Applies to Version 8.33 and later

Laser Process Control Function OPERATOR'S MANUAL

MARUULPRC08181E REV B

ORIGINAL INSTRUCTIONS

Thank you very much for purchasing a FANUC robot.

Before using the robot, be sure to read the, *FANUC Robot series SAFETY HANDBOOK (B-80687EN)* and understand its contents.

- No part of this manual may be reproduced, copied, downloaded, translated into another language, published in any physical or electronic format, or in any other form, including the internet, or transmitted in whole or in part in any way without the prior written consent of FANUC or FANUC America Corporation.
- The appearance and specifications of this product are subject to change without notice.

The products in this manual are controlled based on Japan's "Foreign Exchange and Foreign Trade Law". The export from Japan may be subject to an export license by the government of Japan. Further, re-export to another country may be subject to the license of the government of the country from where the product is re-exported. Furthermore, the product may also be controlled by re-export regulations of the United States government. Should you wish to export or re-export these products, please contact FANUC for advice.

In this manual, we endeavor to include all pertinent matters. There are, however, a very large number of operations that must not or cannot be performed. Please assume that any operations that are not explicitly described as being possible are not possible.

SAFETY PRECAUTIONS

This chapter describes the precautions which must be followed to enable the safe use of the robot. Before using the robot, be sure to read this chapter thoroughly.

For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell. For safe use of FANUC robots, you must read and follow the instructions in the *FANUC Robot series SAFETY HANDBOOK (B-80687EN)*.

PERSONNEL

Personnel can be classified as follows:

Operator:

- Turns the robot controller power on/off
- Starts the robot program from the operator panel

Programmer or Teaching operator:

- Operates the robot
- Teaches the robot inside the safeguarded space

Maintenance technician:

- Operates the robot
- Teaches the robot inside the safeguarded space
- Performs maintenance (repair, adjustment, replacement)

The operator is not allowed to work in the safeguarded space.

The programmer or teaching operator and maintenance technician are allowed to work in the safeguarded space. Work carried out in the safeguarded space include transportation, installation, teaching, adjustment, and maintenance.

To work inside the safeguarded space, the person must be trained on proper robot operation.

[Table s-1](#) lists the work outside the safeguarded space. In this table, the symbol “○” means the work is allowed to be carried out by the specified personnel.

Table s-1 Work Performed Outside the Safeguarded Space

	Operator	Programmer or Teaching Operator	Maintenance Technician
Turn power ON/OFF to Robot controller	○	○	○
Select operating mode (AUTO, T1, T2)		○	○
Select remote/local mode		○	○
Select robot program with teach pendant		○	○
Select robot program with external device		○	○
Start robot program with operator's panel	○	○	○

Start robot program with teach pendant		o	o
Reset alarm with operator's panel		o	o
Reset alarm with teach pendant		o	o
Set data on teach pendant		o	o
Teaching with teach pendant		o	o
Emergency stop with operator's panel	o	o	o
Emergency stop with teach pendant	o	o	o
Operator Panel maintenance			o
Teach Pendant maintenance			o

During robot operation, programming and maintenance, the operator, programmer, teaching operator and maintenance engineer take care of their own safety using at least the following safety protectors:

- Use clothes, uniform, overall adequate for the work
- Safety shoes
- Helmet

DEFINITION OF SAFETY NOTATIONS

To ensure the safety of workers and prevent damage to the machine, this manual indicates each precaution on safety according to its severity. Read the contents of each **Warning** and **Caution** before attempting to use the robot.

WARNING

Indicates a hazard resulting in the death or serious injury of the user could occur if he or she fails to follow the approved procedure.

CAUTION

Indicates a hazard resulting in the injury of the user or damage to equipment could occur if the user fails to follow the approved procedure.

Note

Indicates a supplementary explanation not related to any WARNING or CAUTION.

TABLE OF CONTENTS

SAFETY PRECAUTIONS.....	s-1
1 LASER PROCESS CONTROL FUNCTION (R828).....	1
1.1 FUNCTION OVERVIEW.....	1
1.2 LASER PROCESS CONTROL SOFTWARE VERSION AND PLATFORM.....	1
1.3 ARCTOOL SETUP.....	1
1.4 LIMITATIONS.....	2
2 CONTROL SIGNAL CHANNELS SPECIFICATION.....	3
3 LASER WELDER AND WIRE FEEDER SPECIFICATION.....	5
4 LASER START AND LASER END TIMING SPECIFICATION.....	7
5 UIF.....	9
5.1 ARCTOOL SETUP MENU AT CTRL START.....	9
5.2 LASER WELD SCHEDULE DATA SCREEN.....	9
5.3 LASER TIMING SETUP.....	11
5.3.1 Laser Timing Setup Menu.....	11
5.3.2 Laser Timing Description.....	12
5.3.3 Laser Timing Check.....	13
5.3.4 Laser Welding Start Timing.....	14
5.4 I/O WELD FOR LASER.....	15
5.5 WELD STATUS.....	16
5.6 LASER START AND LASER END INSTRUCTIONS.....	16
6 RAMPING CONTROL.....	19
6.1 RAMP UP CONTROL AT LASER START (\$AWLASER[CHN]. \$RAMP_UP[SCH]).....	19
6.2 RAMP DOWN CONTROL AT LASER END (\$AWLASER[CHN]\$RAMP_DN[SCH]).....	19
6.3 RAMP CONTROL AT SCHEDULE CHANGE (\$AWLASER[CHN].\$RAMP_UP[SCH]).....	20
6.4 RAMP UP EXAMPLES.....	20
6.5 RAMP DOWN EXAMPLES.....	21
7 WIRE RETRACT.....	25
7.1 WIRE RETRACT SETUP.....	25
7.2 WIRE RETRACT TIMING.....	25
7.3 WIRE RETRACT EXAMPLE.....	26
8 RAMP SCHEDULE CHANGE EXAMPLE.....	27
9 PROCESS ADAPTIVE CONTROL.....	31
9.1 LINEAR PROPORTIONAL CONTROL ALGORITHM.....	31
9.2 CONTROL LOOP TIME.....	31
9.3 FILTERING COMMAND CONTROLS.....	32
9.4 MONITORING FEEDBACK VALUE.....	33
10 SYSTEM VARIABLES.....	35
10.1 \$AWLASER[CFG].....	35

10.2 \$AWLASER[CHN]..... 36

11 PROGRAMMING..... 41

11.1 PROGRAM EXAMPLE 1 (SINGLE SCHEDULE WITH CORNER MOVES)..... 41

11.2 PROGRAM EXAMPLE 2 (SCHEDULE CHANGES)..... 42

12 ERROR CONDITIONS AND ERROR RECOVERY..... 45

13 ERROR HANDLING..... 47

13.1 USER DEFINED ERROR CONDITIONS..... 47

13.2 LASER FEEDBACK CHECK..... 47

13.3 WIRE FEED FEEDBACK CHECK..... 47

14 ALARMS..... 49

15 I/O CONNECTIONS..... 51

15.1 MOCK ARC WELD I/O..... 51

15.1.1 Configuring the Memory Image IO and Assigning the Arc Weld IO..... 52

15.2 I/O CONNECTION TO IPG..... 53

15.3 I/O CONNECTION TO LASERLINE..... 54

15.4 I/O CONNECTION TO ABICOR BINZEL WIRE FEEDER..... 55

1 LASER PROCESS CONTROL FUNCTION (R828)

Laser Process Control (LPC) function provides a flexible and accurate timing synchronization control for laser power and wire feed speed for applications of laser braze and laser weld. It further provides a flexible and independent adaptive control for laser power and wire feed speed with respect to actual welding speed.

Note

This manual applies to software versions V8.33P/10 or later versions.

1.1 FUNCTION OVERVIEW

Following functionalities are available for Laser Process Control software option R828 in v8.30 release.

- Stationary Laser Start and Laser End in FINE move
- Flexible ramping controls weld start at acceleration and weld end at deceleration
- Flexible ramping controls during weld at schedule change
- Adaptive control of laser power based on actual weld speed
- Adaptive control of wire feed speed based on actual weld speed
- Independent control ON timing for laser power and wire feed speed at `Laser Start`
- Independent control ON timing for laser on and wire feed on at `Laser Start`
- Independent control OFF timing for laser power and wire feed speed at `Laser End`
- Independent control OFF timing for laser and wire feed at `Laser End`
- Hold/Resume and Original Path Resume for error recovery
- One point of setup and easy for programming
- Up to 32 weld schedules
- Support `Laser Start` and `Laser End` TPE instructions

1.2 LASER PROCESS CONTROL SOFTWARE VERSION AND PLATFORM

Laser Process Control (R828) is a software option in ArcTool.

This function includes J682 Weld Speed output and requires R663 Constant Path (load and enable)

1.3 ARCTOOL SETUP

At control start, select Manufacturer and Model as shown below.

Manufacturer	Laser General
Model	Laser (WATT, WFS) or Laser (WATT)

Multi-process

DISABLED

1.4 LIMITATIONS

The following limitations apply:

- Does not support multiple laser robot groups.
- Does not support coordinated motion or remote TCP motion.

2 CONTROL SIGNAL CHANNELS SPECIFICATION

The Laser Process Control software supports two channels (Laser Power and Wire Feed). Each channel contains seventeen bits of control for commanded output and feedback input respectively. In addition, it also provides three control bits of wire feed backward, inch forward and inch backward for Binzel feeder.

Commanded Output

- Laser ON DO (1 bit)
- Wire Feed ON DO (1 bit)
- Wire Feed Backward DO (1 bit) for Binzel feeder
- Wire Inch Forward DO (1 bit) for Binzel feeder
- Wire Inch Backward DO (1 bit) for Binzel feeder
- Laser Power GO or AO (≤ 16 bits)
- Wire Feed Rate GO or AO (≤ 16 bits)

Feedback Input

- Laser ON DI (1 bit)
- Wire Feed ON DI (1 bit)
- Laser Power GI or AI (≤ 16 bits)
- Wire Feed Rate GI or AI (≤ 16 bits)

3 LASER WELDER AND WIRE FEEDER SPECIFICATION

Supports:

- Abicor Binzel wire feeder.
- IPG Laser, laserline, and TRUMPF TruDisk Laser.

4 LASER START AND LASER END TIMING SPECIFICATION

The chart below specifies signal timing for two channels (Laser & Wire Feed) in FINE move type at Laser Start and Laser End. Each channel has an independent control. The timing specification applies to both weld schedule and weld procedure.

- Either triggers of $\$va_of_dly$ or $\$do_of_dly$ at Laser End will turn off the $\$cmd.\val for each channel
- The $\$mo_dly_as$ is the delay time relative to Feedback Laser On time at Laser Start
- The $\$mo_dly_ae$ is the delay time relative to Command Laser Off time at Laser End
- All ON control must be completed within **Laser Start process period**
- All OFF control must be completed within **Laser End process period**
- The $\$mo_dly_as$ must be less than $\$awsemoff.\chk_time to avoid **ARC-054** error.

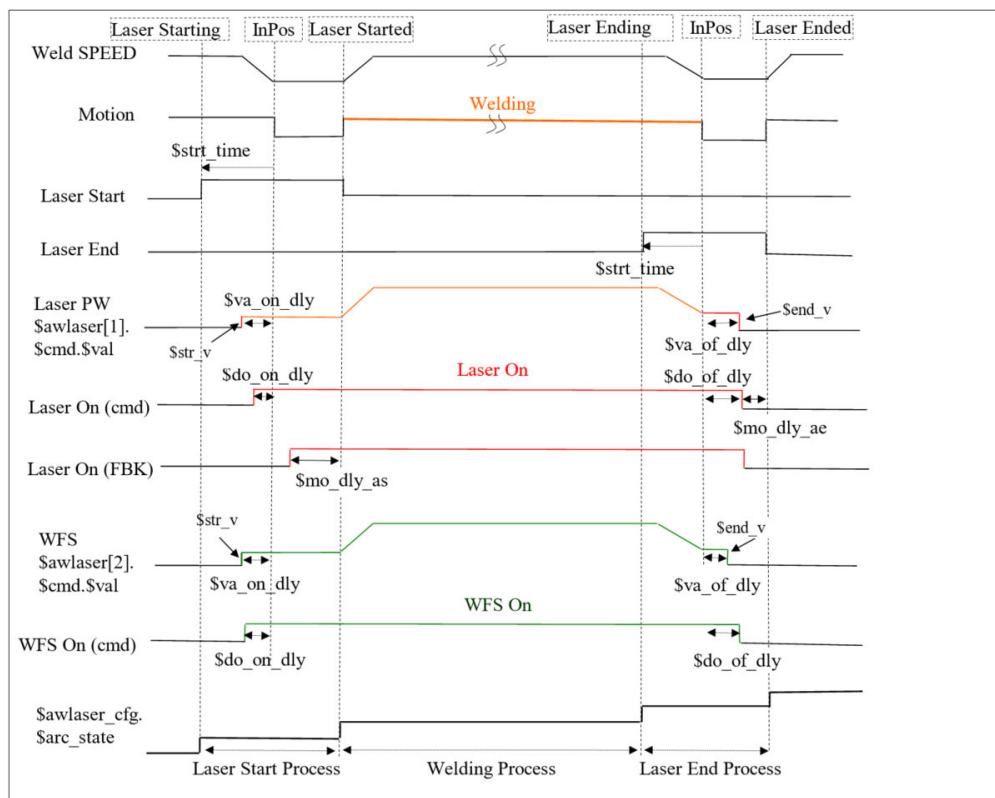


Figure 4 Laser Start and Laser End Timing

5 UIF

5.1 ARCTOOL SETUP MENU AT CTRL START

The setup will set **Laser General** as the default Manufacturer and **Laser (WATT, WFS)** as the default Model at INIT loading if Laser Process Control (R828) is loaded and the laser process control is enabled (*\$awlaser_cfg.\$enable* = TRUE). The screen below shows the Default Laser Setup screen. You can add a user defined KAREL program in the setup item 7 and enable the KAREL program execution in the item 8. Once the specified KAREL program is executed, the Program Execution will be disabled automatically.

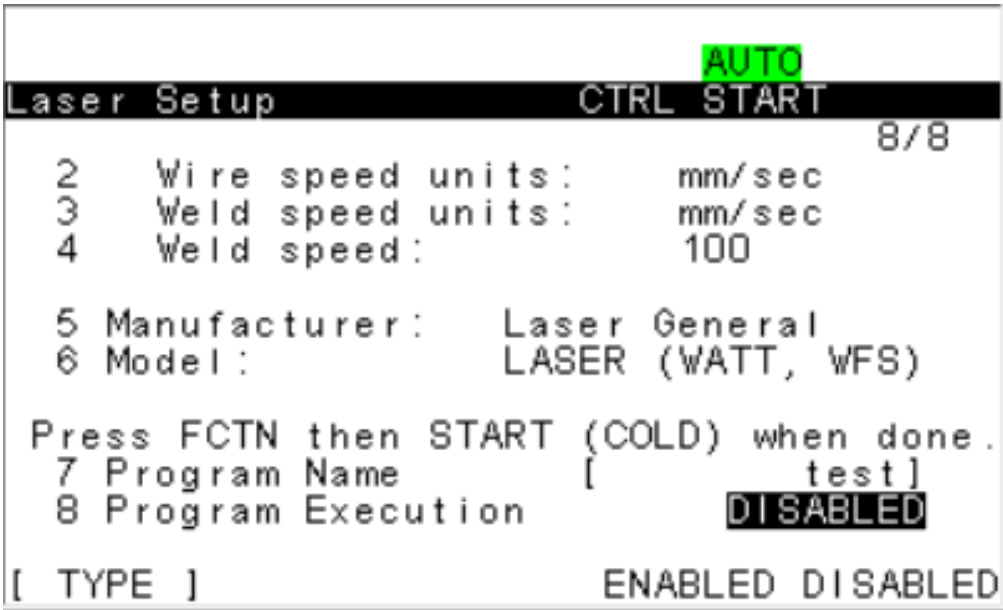


Figure 5.1 Manufacture Setup

5.2 LASER WELD SCHEDULE DATA SCREEN

The Laser Process Control supports 32 welding schedules. Press the **DATA** and then **F1, TYPE**; you can select **Weld Sched** to display the laser weld schedule screen as shown below.

DATA Weld Sched				
1/32				
	WATTS	mm/sec	mm/sec	COMMENT
1	5000	200	40	
2	5000	200	15	
3	5000	200	15	
4	5000	200	15	
5	5000	200	15	
6	5000	200	15	
7	5000	200	15	
8	5000	200	15	
9	5000	200	15	
10	5000	200	15	
LaserPW 5000 WATTS (0, 10000)				
[TYPE] DETAIL TIMING ? HELP >				

Figure 5.2 (a) Laser (PW) Weld Schedule

Press F2, DETAIL the detail schedule screen is shown below.

DATA Weld Sched				
1/18				
1	Weld Schedule	1	[*****]	
2	LaserPW		5000.0	WATTS
3	Wire feed		200.0	mm/sec
4	Travel speed		40.0	mm/sec
-- Laser Parameters --				
5	Adapt gain		1.0	
6	Ramp up mode		USER	
7	Ramp up time		192	msec
8	Ramp up strt val		20	WATTS
9	Ramp up dly time		0	msec
[TYPE] SCHEDULE ? HELP >				

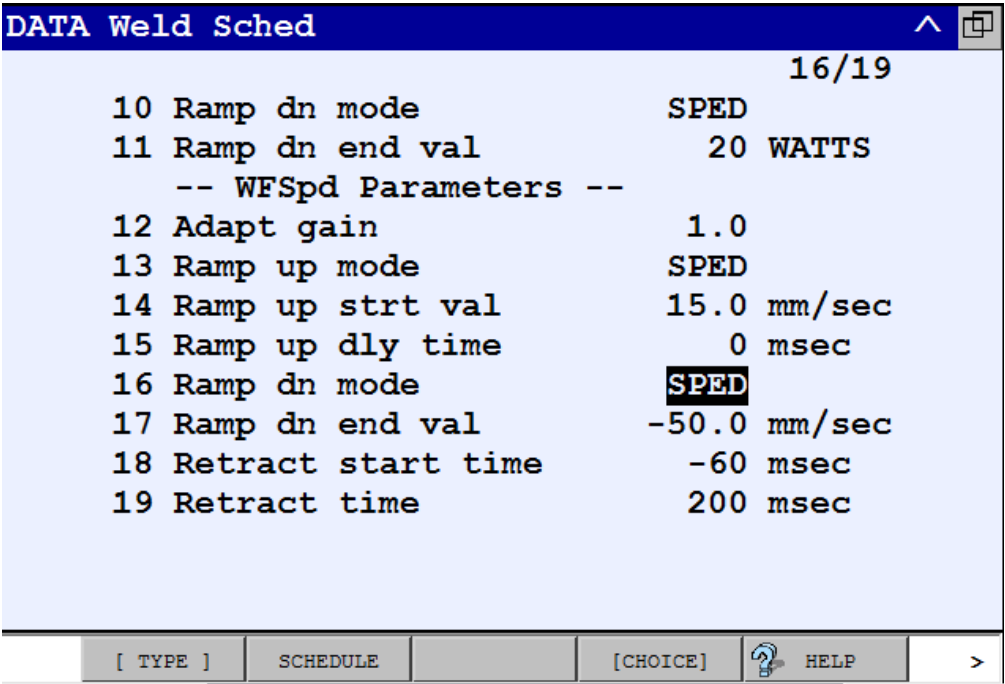


Figure 5.2 (b) Detail Schedule for Laser (PW, WFS)

Note

- The item of **Ramp up time** or **Ramp dn time** will be shown on the screen only if the Ramp mode is USER.
- The items of **Retract start time** and **Retract time** will be shown only if **Ramp dn end val** is less than 0.

5.3 LASER TIMING SETUP

5.3.1 Laser Timing Setup Menu

The laser timing setup menu is attached to the Data Weld Sched screen. Select **F3, TIMING** in DATA Weld Schedule, you will see the Laser Timing Screen.

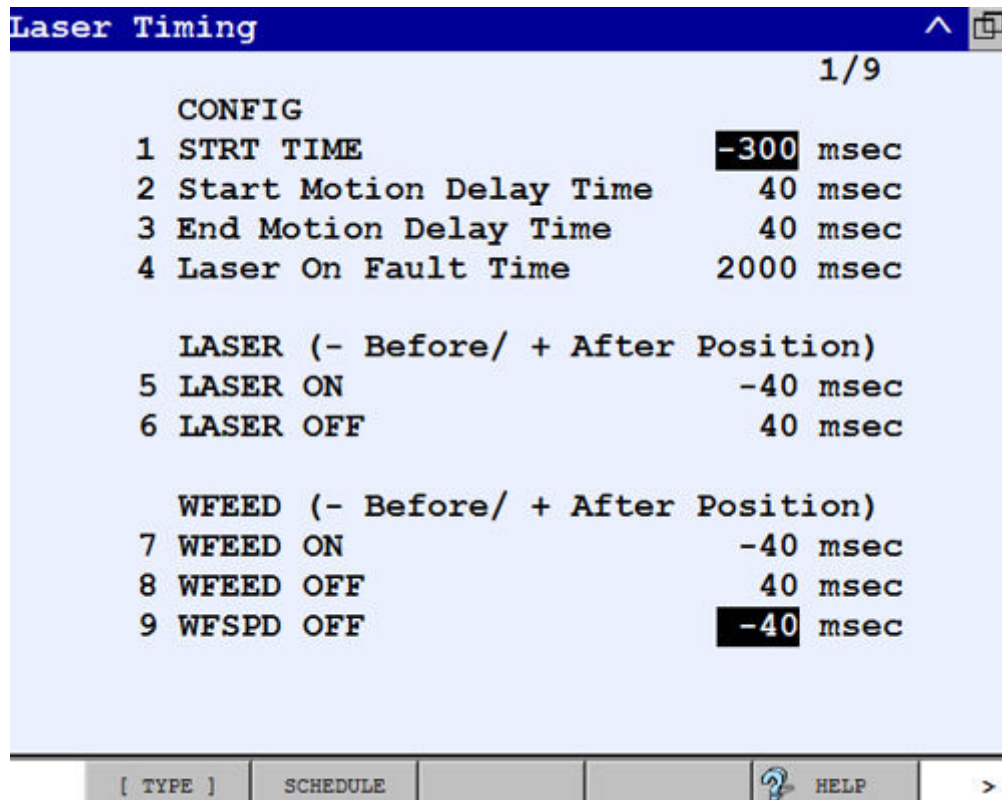


Figure 5.3.1 Laser Timing Screen

5.3.2 Laser Timing Description

Table 5.3.2 Timing Description

Item	Name	Description	Sysvars
1	Start Time	Specifies Laser Starting or Laser Ending time before robot reaches to a taught position that has Laser Start or Laser End instruction.	<i>\$awlaser_cfg.\$strt_time</i>
2	Start Motion Delay Time	Specifies the motion delay time after laser is detected on (fbk) at Laser Start process.	<i>\$awlaser_cfg.\$mo_dly_as</i>
3	End Motion Delay Time	Specifies the motion delay time after laser is off (cmd) at Laser End process.	<i>\$awlaser_cfg.\$mo_dly_ae</i>
4	Laser On Fault Time	Specifies the max motion delay time after laser is on (cmd) at Laser Start process. If the laser is not detected on, Alarm 408 is posted.	<i>\$awlaser_cfg.\$wait_time</i>

5	Laser On	Specifies the laser on time relative to the taught position that has Laser Start instruction.	$\$awlaser[1].\do_on_dly
6	Laser Off	Specifies the laser off time relative to the taught position that has Laser End instruction.	$\$awlaser[1].\do_of_dly
7	WFEED On	Specifies the wire feed on time relative to the taught position that has Laser Start instruction.	$\$awlaser[2].\do_on_dly
8	WFEED Off	Specifies the wire feed off time relative to the taught position that has Laser End instruction.	$\$awlaser[2].\do_of_dly
9	WFSPD Off	Specifies the wire feed analog off time relative to the taught position that has Laser End instruction.	$\$awlaser[2].\va_of_dly

5.3.3 Laser Timing Check

At `Laser Start`, the `Start Time` specifies the low limit. The `Laser On + Start Motion Delay Time` defines the upper limit. Both lower and upper limit are with respect to the time robot reaches to the start welding position. The plot below illustrates the timing window at the Laser Start.

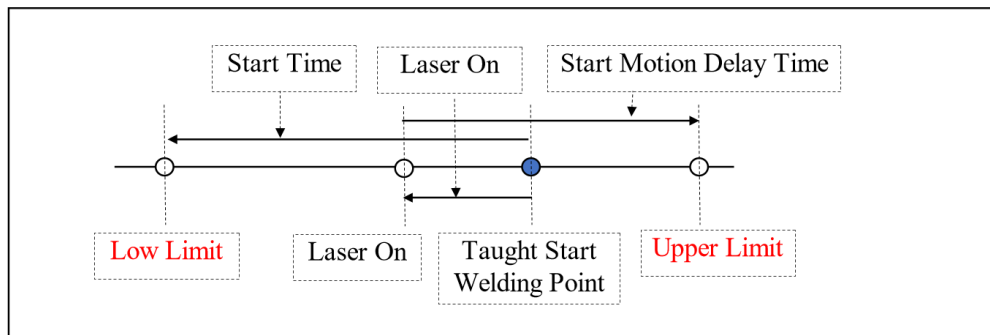


Figure 5.3.3 (a) Laser Timing Window at Laser Start

At `Laser End`, the `Start Time` specifies the low limit. The `Laser Off + End Motion Delay Time` defines the upper limit. Both lower and upper limit are with respect to the time robot reaches to the end welding position. The plot below illustrates the timing window at the Laser End.

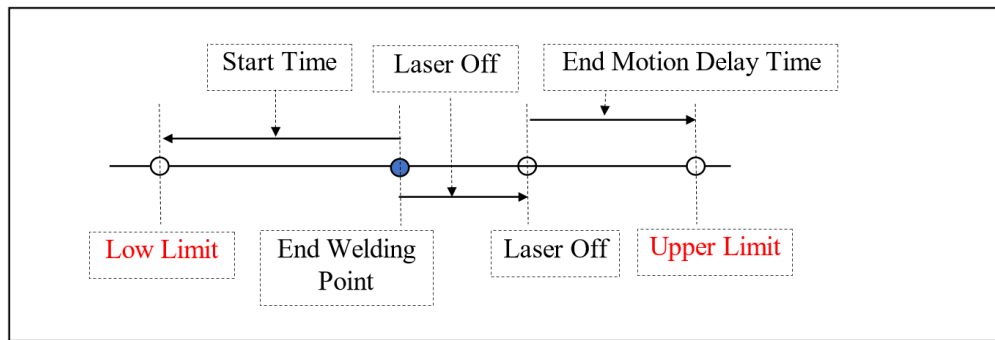


Figure 5.3.3 (b) Laser Timing Window at Laser End

The start time, the laser on time, and the Start motion delay time define the low and upper time limit at the Laser Start as illustrated in Figure 5.3.3 (a). The start time, the laser off time, and the End motion delay time define the low and upper time limit at the Laser End as illustrated in Figure 5.3.3 (b). These timing parameters are independently determined. Once these parameters are defined, the dependent timing parameters must be defined accordingly. For example, all the wire feed timing parameters will be checked in relative to the defined timing window. If the set value is out of the window, the system will prompt the message indicating if this is out of lower or upper limit.

5.3.4 Laser Welding Start Timing

The laser welding motion start timing is based on the feedback detected laser on time plus a user defined Start Motion Delay Time. The maximum delay time is specified by the command laser on time plus a user defined Laser On Fault Time. Normally, the Start Motion Delay Time occurs before the Laser On Fault Time. However, if the Fault Time occurs prior to the Start Motion Delay Time and the laser is detected on, the laser welding motion will start right way. If the laser is NOT detected at the Laser On Fault Time, the system will post an **ARC-408** error.

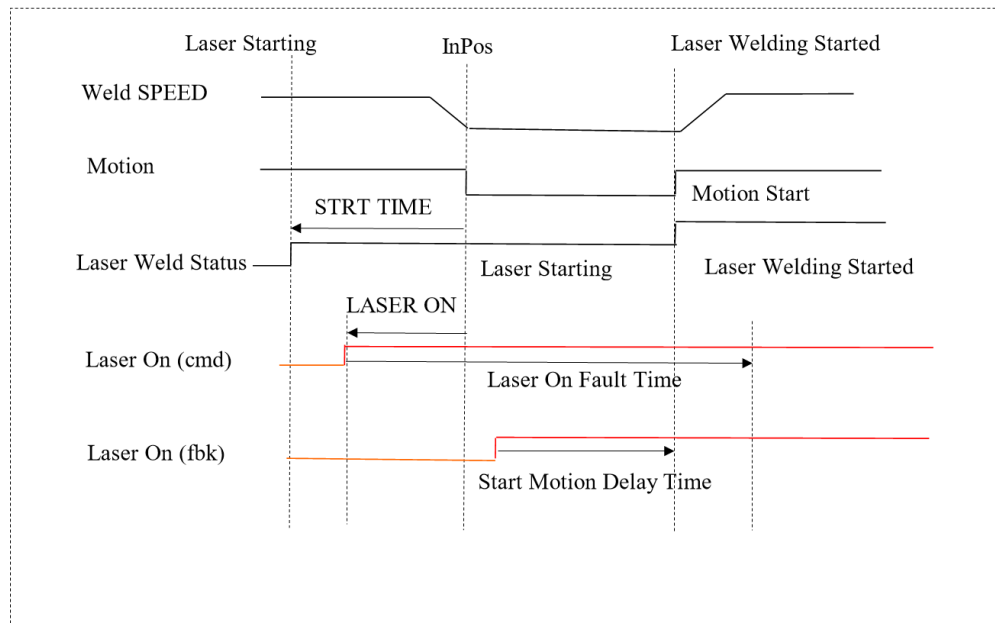


Figure 5.3.4 Laser Welding Start Timing

5.4 I/O WELD FOR LASER

The Weld I/O setup will allow you to setup the command and feedback I/O for laser and wire channels. You can get the I/O weld setup screen by selecting **MENU > I/O > Weld**. You can toggle the input and output by pressing **F3, IN/OUT** while in the I/O weld display screen.

I/O Weld Out					
1/7					
	WELD SIGNAL	TYPE #	SIM	STATUS	
1	[Cmd Laser Pwr]	GO[1]	U	0	
2	[Cmd Wire Speed]	GO[2]	U	0	
3	[Laser On	DO[1]	U	OFF	
4	[Wire forward]	DO[2]	U	OFF	
5	[Wire backward]	DO[3]	U	OFF	
6	[Inch forward]	DO[4]	U	OFF	
7	[Inch backward]	DO[5]	U	OFF	

[TYPE]

HELP

IN/OUT

Figure 5.4 (a) Laser I/O Out Screen

I/O Weld In					
1/4					
	WELD SIGNAL	TYPE #	SIM	STATUS	
1	[Act Laser Power]	GI[1]	U	0	
2	[Act Wire Speed]	GI[2]	U	0	
3	[Laser On	DI[1]	U	OFF	
4	[Wire feed On	DI[2]	U	OFF	

[TYPE]

HELP

IN/OUT

Figure 5.4 (b) Laser I/O In Screen

Note

The laser weld I/O display is the default. If you want to view the regular ARC weld I/O screens you can disable the laser process control by setting `$awlaser_cfg.$enable = FALSE`.

5.5 WELD STATUS

A **Laser Status** screen provides you the **Command** and actual **Feedback** values of the laser power, wire feed speed, as well as the laser and wire status in real time. You can view the **Laser Status** screen by selecting **MENU > STATUS > Weld**.

Laser Status			
	Command	Feedback	
Laser Power	4992	4992	WATTS
Wire Feed	200	200	mm/sec
Weld Speed	40	40	mm/sec
Laser On	ON	ON	

Figure 5.5 Laser Weld Status Screen

5.6 LASER START AND LASER END INSTRUCTIONS

When laser process control is loaded and enabled, the `Laser Start` and `Laser End` instructions will replace `Arc Start` and `Arc End` respectively.

You can attach a Laser instruction to a move point from **Motion Modify** as shown below.

Motion Modify		1
1	No option	
2	Laser Start ()	
3	Laser End()	
4	Wrist Joint	
5	ACC	
6	Skip, LBL	
7	BREAK	

Figure 5.6 Laser Instruction

6 RAMPING CONTROL

The ramp control plays important roles in achieving a good quality of laser braze and laser weld at `Laser Start`, `Laser End` or at the schedule change. This function provides four ramp modes that allow you to control the channel parameter ramping in various ways during motion acceleration at `Laser Start` and deceleration at `Laser End` or at `schedule change` during the welding. Different channel can use different ramp mode. The ramp can be controlled independently for each channel and for each schedule.

6.1 RAMP UP CONTROL AT LASER START (\$AWLASER[CHN]. \$RAMP_UP[SCH])

The ramp up schedule number is specified by `Laser Start [sch]`. The ramp up provides following controls:

- None Ramp ($\$mode = 1$). In this mode, the process value will start at command value and maintain its value until acceleration completes.
- Auto Ramp ($\$mode = 2$). In this mode, the ramp time is the motion acceleration time. The ramp start timing is synchronized with the motion start timing. The process value will ramp from any specified start ramp value to the command value linearly in the acceleration time.
- User Ramp ($\$mode = 3$). In this mode, user can specify a desired ramp up time ($\geq 16\text{ms}$). The ramp start timing is synchronized with the motion start timing. The process value will ramp from any specified start ramp value to the command value linearly in the ramp up time.
- SPED Ramp ($\$mode = 4$). In this mode, the ramp time is the motion acceleration time. The ramp start timing is synchronized with the motion start timing. The process value will ramp from any specified start ramp value to command value in accordance to the actual speed profile during acceleration.
- The $\$str_v$ specifies the ramp start value. This is absolute value.
- The $\$dly_t$ specifies the ramp delay time in milliseconds. It delays the ramp start time after motion starts. This delay time can be applied to ramp up modes of AUTO, USER and SPED.

6.2 RAMP DOWN CONTROL AT LASER END (\$AWLASER[CHN]\$RAMP_DN[SCH])

The ramp down schedule number is specified by `Laser Start [sch]`. The schedule number in `Laser End` is not used. The ramp down provides following controls:

- None Ramp ($\$mode = 1$). In this mode, the process value is maintained to its last adaptive control value during deceleration period, and will become zero when either of the digital or analog signal is off.
- Auto Ramp ($\$mode = 2$). In this mode, the ramp time is the motion deceleration time. The ramp down start timing is synchronized with the motion deceleration start timing. The process value will ramp down from current value to the specified end ramp value linearly in the deceleration time.
- User Ramp ($\$mode = 3$). In this mode, user specifies the ramp down time ($\geq 16\text{ms}$). The ramp down time is relative to the time at the weld stop position. The process value will ramp down from current value to the specified end value linearly. The ramp ends at the time when robot stops. When the digital control signal or the analog signal is shut off prior to the robot stop, the ramp will end immediately.

- SPED Ramp ($\$mode = 4$). In this mode, the ramp time is the motion deceleration time. The ramp starts at current value and ends at the specified end value. The ramp is in accordance with the speed profile during deceleration.
- When $\$end_v \geq 0$, it specifies command end value at ramp down. This is absolute value.
- When $\$end_v < 0$, it specifies wire retract speed. Negative ramp end value can only be applied to wire feed channel.

6.3 RAMP CONTROL AT SCHEDULE CHANGE (\$AWLASER[CHN].\$RAMP_UP[SCH])

When schedule is changed during weld, the process (laser power and wire feed speed) transition is controlled by ramp up schedule of the new schedule. The ramp control at schedule change provides following controls:

- None Ramp ($\$mode = 1$). In this mode, the process value will be changed from its last adaptive control value to the new command value immediately at the schedule change.
- Auto Ramp ($\$mode = 2$). In this mode, the ramp time is the motion transition time. The ramp starts at current value and is ramping to the new command value linearly.
- User Ramp ($\$mode = 3$). In this mode, user specifies the ramp time. If user defined ramp time is less than the motion transition time, the system will automatically set the motion transition time for the ramp time. The ramp starts at current value and is ramping to the new command value linearly.
- SPED Ramp ($\$mode = 4$). In this mode, the ramp time is the motion transition time. The ramp starts at current value and is ramping to the new command value in accordance with the speed profile.
- When weld speed in new schedule is the same as to the previous weld speed, the system will automatically switch to the Auto ramp mode.

6.4 RAMP UP EXAMPLES

The plots below illustrates the ramp up effect in four different modes for the Laser Power channel. In these examples, the laser ramp up start value is 30 and the wire feed start value is 25.

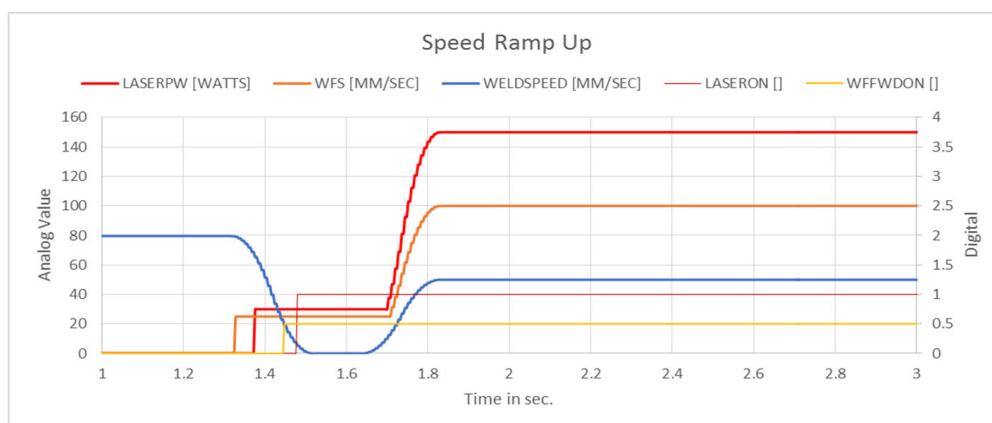


Figure 6.4 (a) Speed Profile Ramp Up

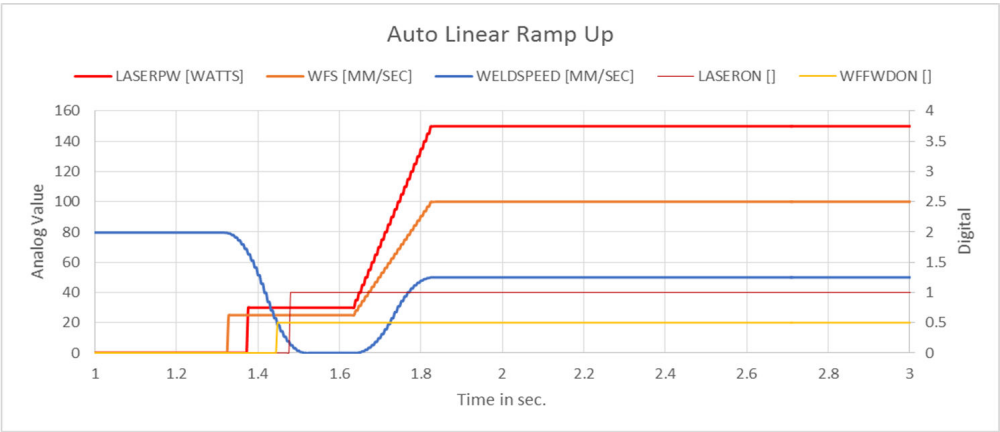


Figure 6.4 (b) AUTO Linear Ramp Up

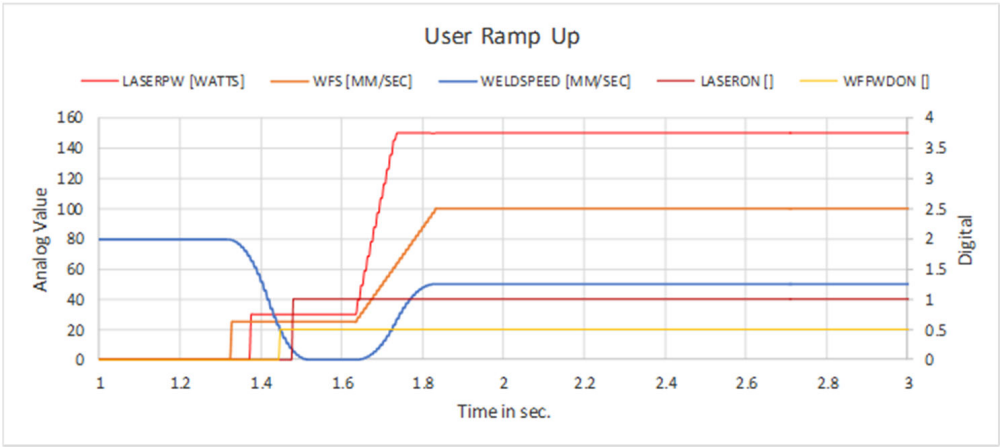


Figure 6.4 (c) USER Defined Linear Ramp Up with 100 ms for laser and 200 ms for wire feed

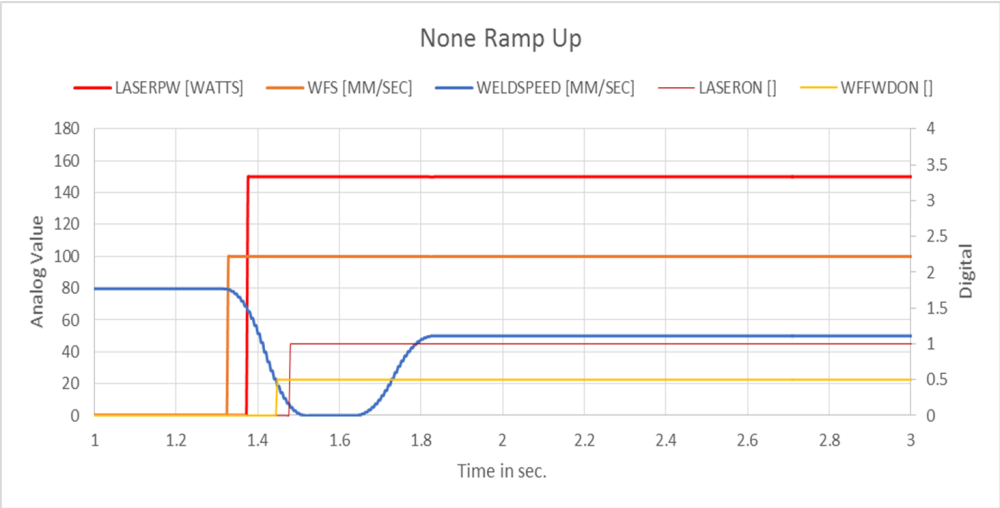


Figure 6.4 (d) None Ramp Up

6.5 RAMP DOWN EXAMPLES

The plots below illustrate ramp down effects in four different modes for Laser Power and wire feed. In these examples, the ramp end value is 30 for laser power and 20 for wire feed.

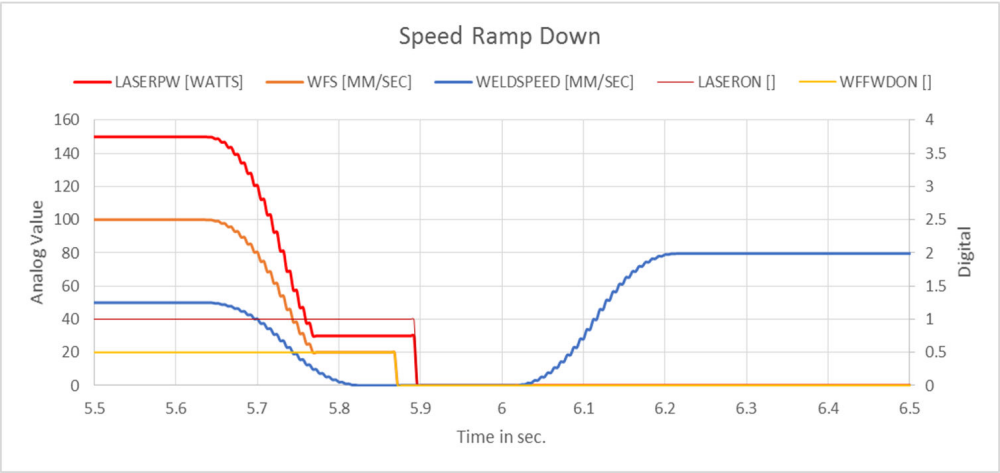


Figure 6.5 (a) Speed Ramp Down

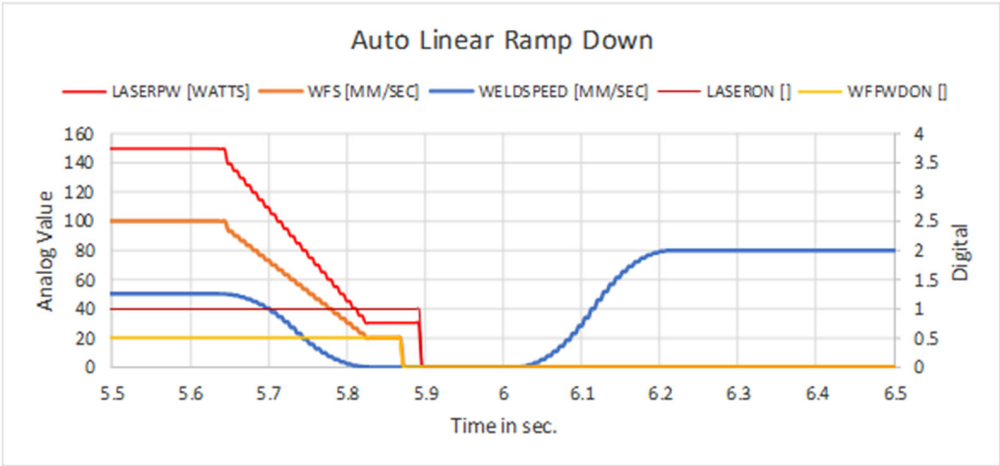


Figure 6.5 (b) Auto Linear Ramp Down

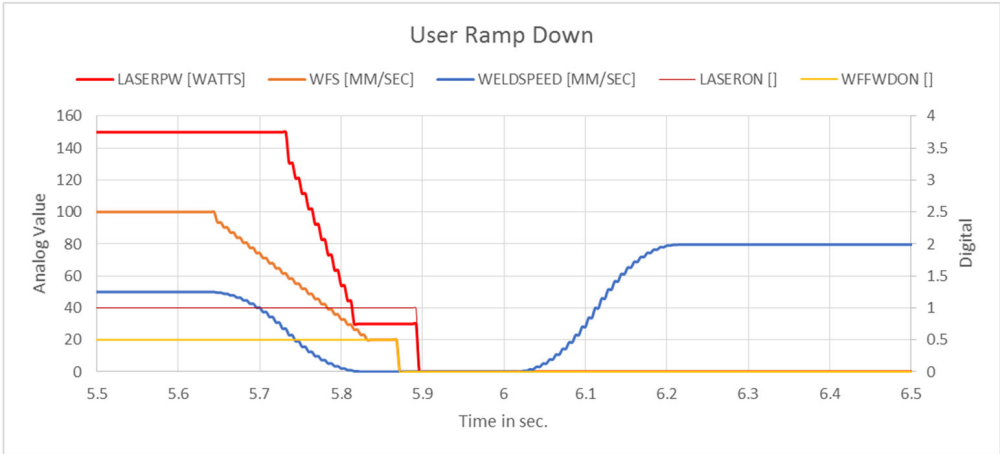


Figure 6.5 (c) User Linear Ramp Down

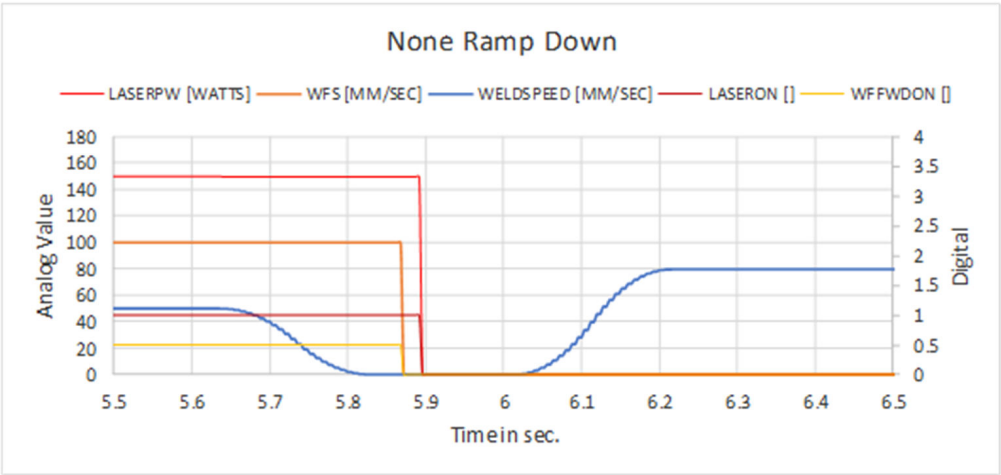


Figure 6.5 (d) None Ramp Down

7 WIRE RETRACT

Wire retract prevents wire stick in the weld pool at the end of the laser braze. Wire retract control is independent of the laser timing control and it is also independent of the ramp down process.

7.1 WIRE RETRACT SETUP

The Wire Retract is specified in the laser weld schedule. A negative value of **Ramp dn end val** (item 16) specifies the wire retract speed (wire backward speed) and enables wire retract operation for this schedule. The **Retract start time** (item 17) specifies the wire retract start time and the **Retract time** (item 18) specifies the retract time. Wire retract supports Binzel wire feeder.

DATA Weld Sched		18/18
9 Ramp dn mode	SPED	
10 Ramp dn end val	20 WATTS	
-- WFSpd Parameters --		
11 Adapt gain	1.0	
12 Ramp up mode	SPED	
13 Ramp up strt val	15.0 mm/sec	
14 Ramp up dly time	0 msec	
15 Ramp dn mode	SPED	
16 Ramp dn end val	-50.0 mm/sec	
17 Retract start time	-60 msec	
18 Retract time	80 msec	

[TYPE]
SCHEDULE
?
HELP
>

Figure 7.1 Wire Retract Setup

7.2 WIRE RETRACT TIMING

At the beginning of wire retract, the WFS forward DO is turned off and the wire backward DO is turned on. At the end of wire retract the wire backward DO is turned off. The retract operation must start within the retract start window and must end before the Laser Ended. (See [Figure 7.2](#)) If the retract time is out of the operation window, the system will end the retract operation at the Laser End time.

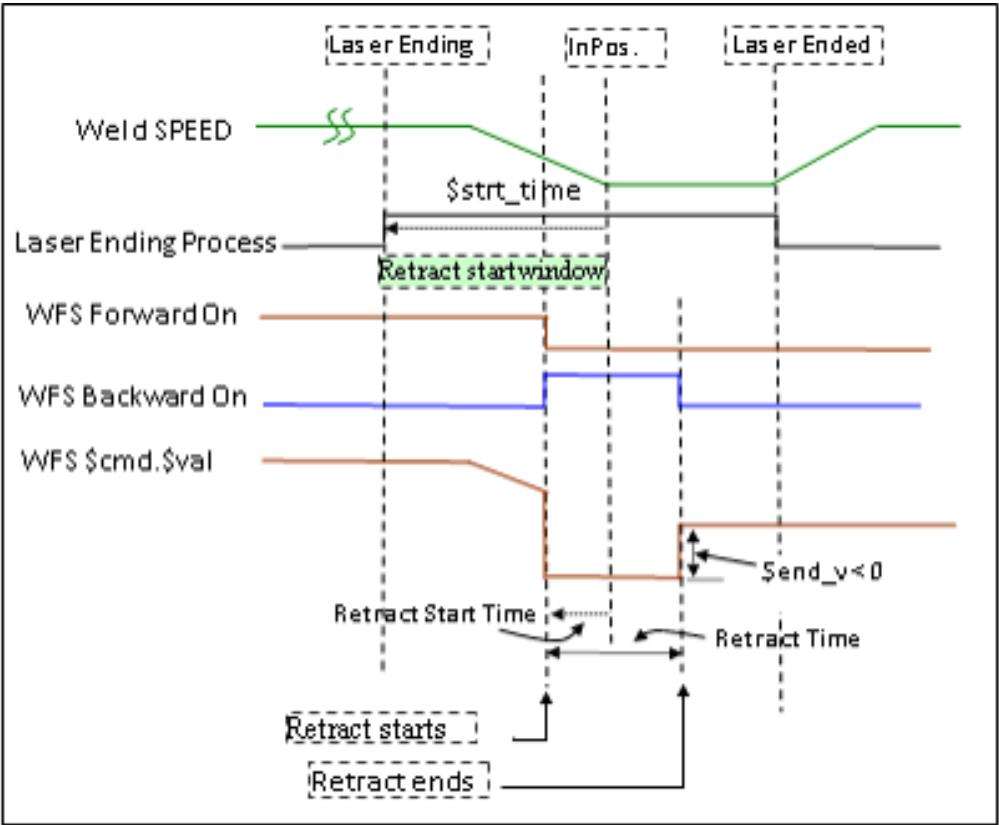


Figure 7.2 Binzel Wire Retract Timing at Laser End

7.3 WIRE RETRACT EXAMPLE

The example below shows the retract effect where the Ramp down end value is -50mm/sec, the retract start time is -40ms and the retract time is 100ms.

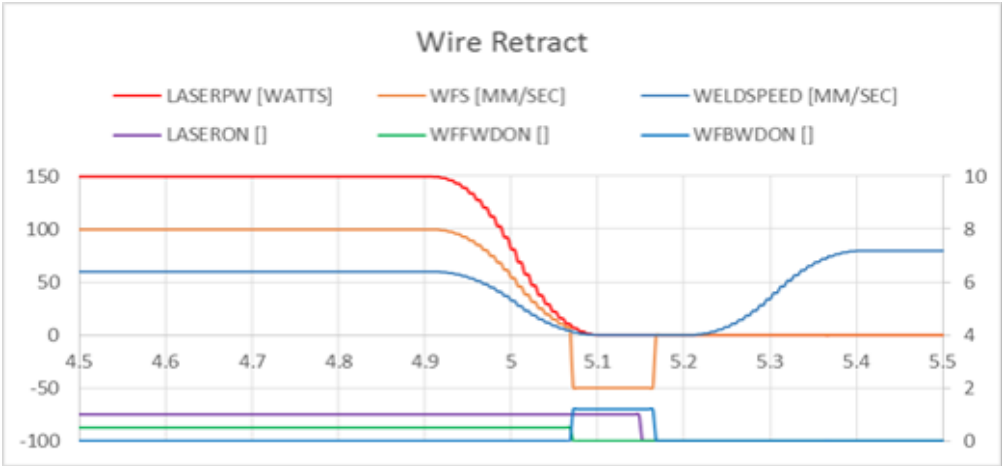


Figure 7.3 Wire Retract Example

8 RAMP SCHEDULE CHANGE EXAMPLE

At schedule change, the ramp up mode of the new schedule is used to control the ramp and its ramp up start value ($\$str_v$) is ignored. When weld speed of new schedule is same as the previous one, the system will automatically use AUTO linear ramp mode.

The plots below illustrate ramp effects at schedule change in four different modes for Laser and WFS channels. The original schedule has the value of LaserPW = 50 WATTS, WFS = 70 mm/sec, Weld SPEED = 60 mm/sec, and the new schedule has the value of LaserPW = 40 WATTS, WFS = 55 mm/sec and Weld SPEED = 45 mm/sec.

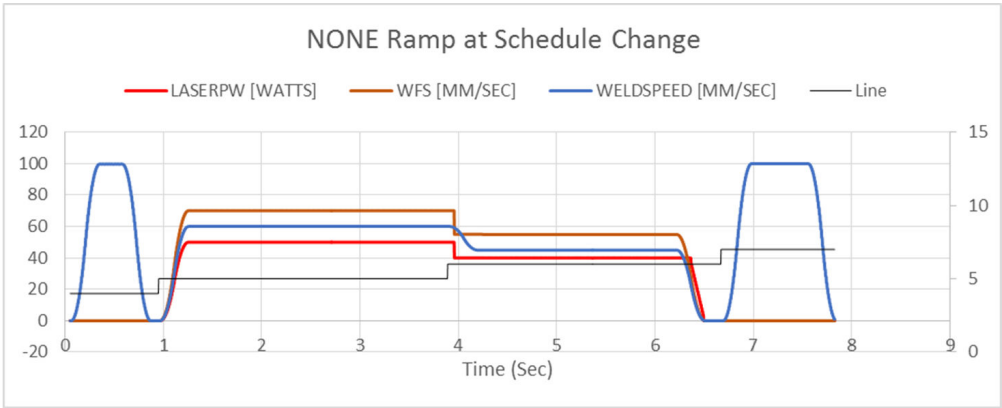


Figure 8 (a) NONE Ramp at Schedule Change

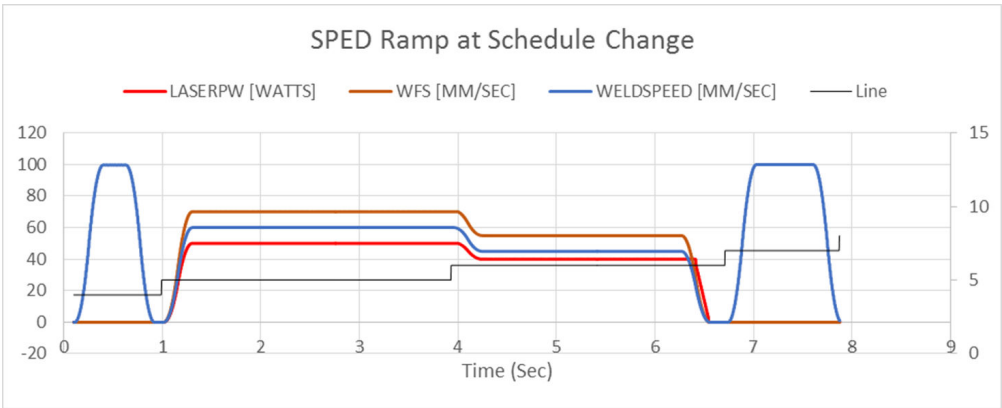


Figure 8 (b) SPED Ramp at Schedule Change

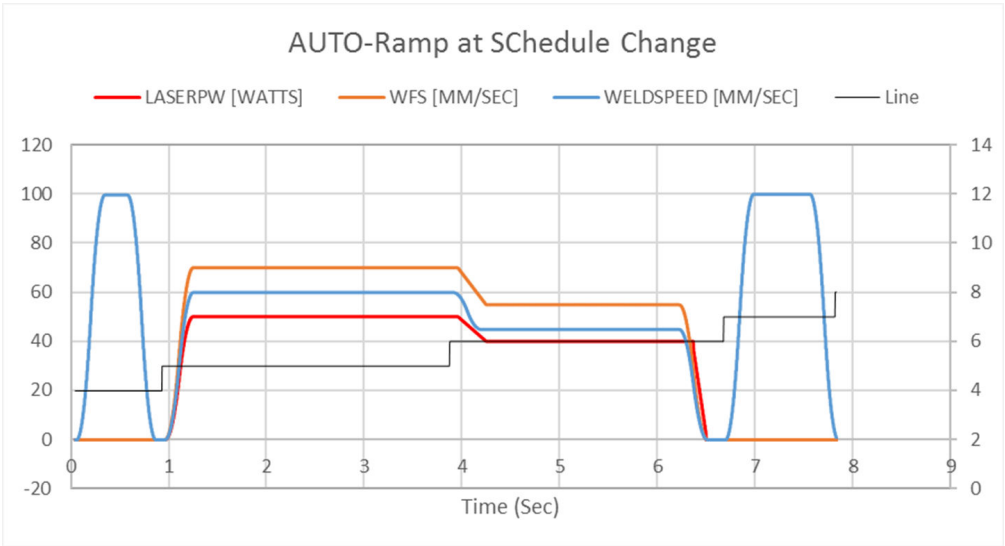


Figure 8 (c) AUTO Ramp at Schedule Change

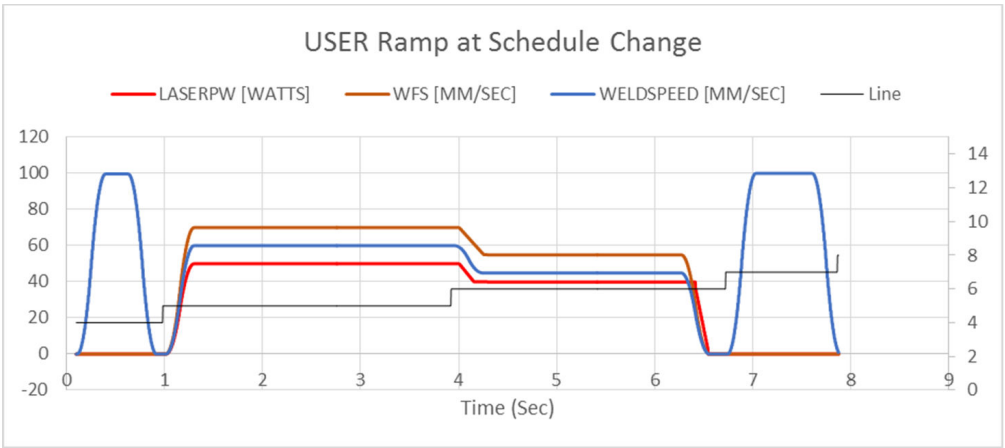


Figure 8 (d) USER Ramp at Schedule Change

In this example, the ramp time is 160ms for Laser and is 250ms for WFS.

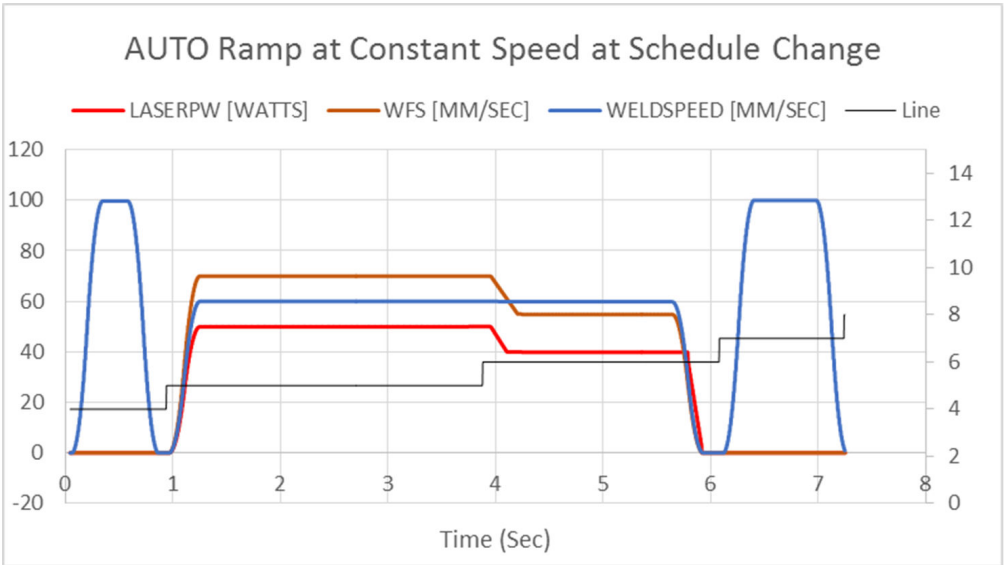


Figure 8 (e) AUTO Ramp at Constant Speed at Schedule Change

In this example, the weld speed for the new schedule is the same as the previous one. The system will automatically use AUTO linear ramp mode.

9 PROCESS ADAPTIVE CONTROL

There are many adaptive control algorithms that can correlate laser power and wire feed speed to actual weld speed. The algorithm below provides a simplest proportional control method.

9.1 LINEAR PROPORTIONAL CONTROL ALGORITHM

Table 9.1 Linear Adaptive Control Algorithm

$\text{speed_ratio} = \text{act_speed} / \text{cmd_speed}$	where $0 \leq \text{speed_ratio} \leq 1$
$\text{lpw} = \text{cmd_lpw} (1 - k1 * (1 - \text{speed_ratio}))$	where $0 \leq k1 \leq 1$ and $\text{lpwmin} \leq \text{lpw} \leq \text{cmd_lpw}$
$\text{wfs} = \text{cmd_wfs} (1 - k2 * (1 - \text{speed_ratio}))$	where $0 \leq k2 \leq 1$ and $\text{wfsmin} \leq \text{wfs} \leq \text{cmd_wfs}$

- The *cmd_speed* is the command weld speed specified in weld schedule
- The *act_speed* is the measured weld speed
- The *speed_ratio* is limited to less than 1.0
- The *cmd_lp* is the command laser power specified in weld schedule
- The *lp* is the adjusted laser power
- The *wfs* is the adjusted wire feed speed
- The *k1* and *k2* are the control gains for *lp* and *wfs* respectively. The changes to these values will be effective until next *Laser Start*.
- The *lpwmin* is the user defined low bound for adjusted laser power
- The *wfsmin* is the user defined low bound for adjusted wire feed

The control gain is specified by channel and schedule variables as

- $k1 = \$\text{awlaser}[1].\$gain[sch]$, for laser power adaptive control
- $k2 = \$\text{awlaser}[2].\$gain[sch]$, for wire feed speed adaptive control

If the control gain is zero or the actual weld speed equals to command weld speed, there is no adjustment to the process parameters. If actual weld speed is slower than the command speed, the process parameters will be reduced proportionally. The gain is used to control the reduction rate and can be selected independently per channel and schedule. You can also disable adaptive control for a specific channel by setting $\$awlaser[channel].\use_adpt to FALSE. When adaptive control is disabled, its control value will follow the command value.

Note

The adaptive control is independent of the ramp control and disabling adaptive control will not automatically disable the ramp control.

9.2 CONTROL LOOP TIME

The control loop time specifies the control granularity (interval). You can specify the control loop time by $\$awlaser_cfg.\$loop_time$, and the system will automatically round the number into a minimum multiple of ITP time during power on of the controller. The minimum control loop time is one ITP

(*\$scr.\$itp_time*). The adaptive control of the laser power and the wire feed speed is updated at control loop interval.

9.3 FILTERING COMMAND CONTROLS

The updated control values at each control loop will be filtered before being applied to the control. For each channel, the maximum allowed incremental change at each control interval is limited by its incremental limit (max or min) and its total value is limited by upper or lower limit. You can set *\$awlaser[CHN].\$cmd.\$chk_lim* to activate the filter control per each channel as:

- Bit0: set this bit to enable the filter control for the lower limit;
- Bit1: set this bit to enable the filter control for the upper limit;
- Bit2: set this bit to enable the filter control for the minimum incremental change;
- Bit3: set this bit to enable the filter control for the maximum incremental change;

If you select to filter the control you must set the filter parameters properly at *\$awlaser[CHN].\$cmd*:

- *\$min_inc*; threshold for the absolute incremental change to be considered per control loop
- *\$max_inc*; threshold for the maximum allowed incremental change per control loop
- *\$min_lim*; minimum allowed control value below the command value
- *\$max_lim*; maximum allowed control value above the command value

Without loss of generality, let $u(t-1)$ be a actual control value at previous control step and $d_u(t)$ be its incremental change at current control step. The $u(t)$ is the control at current step and $u(t) = u(t-1) + d_u(t)$, where the incremental change $d_u(t)$ and the absolute value of $u(t)$ satisfy following constrains:

Table 9.3 (a) incremental Threshold Control

$d_u(t) =$	0	if $\text{abs}(d_u(t)) < \text{min_inc}$
	$d_u(t)$	if $\text{min_inc} < \text{abs}(d_u(t)) < \text{max_inc}$
	max_inc	if $d_u(t) > \text{max_inc}$
	$-\text{max_inc}$	if $d_u(t) < -\text{max_inc}$

Table 9.3 (b) incremental Threshold Control

$d_u(t) =$	0	if $\text{abs}(d_u(t)) < \text{min_inc}$
	$d_u(t)$	if $\text{min_inc} < \text{abs}(d_u(t)) < \text{max_inc}$
	max_inc	if $d_u(t) > \text{max_inc}$
	$-\text{max_inc}$	if $d_u(t) < -\text{max_inc}$

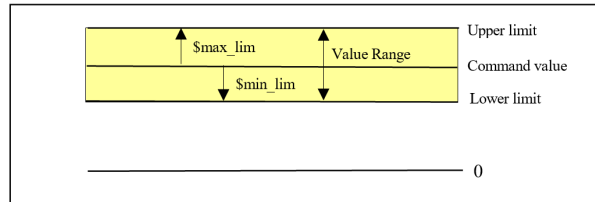
Table 9.3 (c) Incremental Threshold Control

$d_u(t) =$	0	if $\text{abs}(d_u(t)) < \text{min_inc}$
	$d_u(t)$	if $\text{min_inc} < \text{abs}(d_u(t)) < \text{max_inc}$
	max_inc	if $d_u(t) > \text{max_inc}$
	$-\text{max_inc}$	if $d_u(t) < -\text{max_inc}$

Table 9.3 (d) Absolute Value Control

$u(t) =$	$u(t)$	if $\text{cmd_val} + \text{max_lim} < u(t) < \text{cmd_val} - \text{min_lim}$
	$\text{cmd_val} - \text{min_lim}$	if $u(t) < \text{cmd_val} - \text{min_lim}$
	$\text{cmd_val} + \text{max_lim}$	if $u(t) > \text{cmd_val} + \text{max_lim}$

With this control, the commanded value during adaptive control is bounded by its upper and lower limits where these limits are composed by the command value and the channel parameter $\$max_lim$ and $\$min_lim$ at $\$awlaser[CHN].\cmd as

**Figure 9.3 Value Range Window**

The filtered command value for each control channel (laser power or wire feed speed) is updated to its channel system variable of $\$awlaser[chn].\$cmd.\$val$ at control loop interval. The output signal for controlling the weld equipment is further scaled to the range of weld equipment (laser power generator or wire feeder) as

Table 9.3 (e) Output Scaling

$\text{Output} = \$cmd.\$val * \$cmd.\$slop + \$cmd.\$offs$

9.4 MONITORING FEEDBACK VALUE

You can monitor actual feedback values in real time. If a feedback value is out of a user defined window, a warning message is posted. You can set $\$awlaser[CHN].\$fbk.\$chk_lim$ to activate the monitoring function as:

- Bit0: set this bit to monitor lower limit;
- Bit1: set this bit to monitor upper limit;

The window range for the upper and lower limits are composed by the command value and the $\$min_lim$ and $\$max_lim$ of $\$awlaser[CHN].\fbk as

- Lower limit = command value - $\$min_lim$
- Upper limit = command value + $\$max_lim$

You must connect the feedback I/O to the device (Laser power generator or wire feeder) correctly and scale the feedback signal (Group or Analog input) from the weld equipment by

Table 9.4 Input Scaling

$\$fbk.\$val = (\text{feedback Input} - \$fbk.\$offs) / \$fbk.\$slop$

Once the feedback I/O connection is made and $\$awlaser[CHN].\$fbk.\$val$ is scaled correctly, the system can monitor the feedback value at each control loop and will post warning message if the value is out of the window:

- ARC -400 Laser PW Up Limit (TEST, 5) , if $\$awlaser[1].\$fbk\$val > \text{Upper limit for laser channel}$
- ARC -401 Laser PW Lo Limit (TEST, 5) , if $\$awlaser[1].\$fbk\$val < \text{Lower limit for laser channel}$
- ARC -402 WF speed Up Limit (TEST, 5) , if $\$awlaser[2].\$fbk\$val > \text{Upper limit for WFS channel}$
- ARC -403 WF speed Lo Limit (TEST, 5) , if $\$awlaser[2].\$fbk\$val < \text{Lower limit for WFS channel}$

Note

If feedback I/O connection is not set, for example, $\$fbk.\$ou_idx = 0$, system will ignore the monitoring request.

10 SYSTEM VARIABLES

10.1 \$AWLASER[CFG]

Table 10.1 \$awlaser[CHN]

\$awlaser_cfg	Description
<i>\$enable</i> Default: 0	TRUE: enable Laser Process Control function. FALSE: disable function. Change takes effect after re-power of controller.
<i>\$cur_sch</i>	This item indicates current schedule number.
<i>\$loop_time</i> Default: 8 msec	This item specifies the control loop time. Automatically round to minimum of multiple of ITP (<i>\$scr.\$itp_time</i>). Change takes effect after re-power the controller.
<i>\$strt_time</i> Default -300 msec	This item specifies Laser Starting or Laser Ending time before robot reaches to a taught position that has Laser Start or Laser End instruction.
<i>\$wait_time</i> Default 2000 msec	This item specifies Laser On Fault Time in relative to the Laser On Time (cmd). If the laser on is not detected at the Laser On Fault Time, system will post ARC-408 Alarm.
<i>\$mo_dly_as</i> Default : 40 msec	This item specifies motion delay time after laser is detected on (fbk) at Laser Start process.
<i>\$mo_dly_ae</i> Default : 40 msec	This item specifies motion delay time after laser is off (cmd) at Laser End process.
<i>\$fw_do_idx</i>	This item specifies DO index for forward inch wire
<i>\$bw_do_idx</i>	This item specifies DO index for backward inch wire
<i>\$fbw_do_idx</i>	This item specifies DO index for wire feed backward
<i>\$st_io_idx</i>	This item specifies DI index for user defined input signal to stop weld process. If <i>\$st_id_idx</i> = 0, system will not take any action. If <i>\$st_id_idx</i> > 0 and DI value is OFF during the weld, system will stop weld process and post alarm ARC-405 CTRL STOP.
<i>\$spd_schg</i>	This item records speed change (mm/sec) at the schedule change.

\$awlaser_cfg	Description
<i>\$arc_state</i> (read only)	System sets the state: 0: Nothing 1: Laser Starting 2: Laser Started (Weld motion starts) 3: Laser Ending 4: Laser Ended (None-weld motion starts)
<i>\$comp_sw</i>	Internal use
<i>\$debug</i>	Internal use
<i>\$err_flag1</i> Default: 0	User defined error flag. If <i>\$err_flag1</i> > 0, system will post ARC-405 CTRL STOP alarm
<i>\$err_flag2</i> Default: 0	User defined error flag for future use.
<i>\$fname</i>	User define KAREL file name
<i>\$fexec</i>	When enabled, the user defined KAREL file will be executed at CTRL START and is set to FALSE automatically.

10.2 \$AWLASER[CHN]

Table 10.2 \$awlaser[CHN]

\$awlaser[CHN]	Description
<i>\$desc</i>	• Laser for channel 1. • WFS for channel 2.
<i>\$enbl</i> Default: 0	True: enable channel control. False: disable channel control.
<i>\$use_adapt</i> Default: 1	True: enable adaptive. False: disable adaptive.
<i>\$do_on_dly</i> Default: -40 ms	DO ON delay time in msec w/r to a taught point at Laser Start. A negative value refers to the time before and a positive value refers to the time after.
<i>\$do_of_dly</i> Default: 40 ms	DO OFF delay time in msec w/r to a taught point at Laser End. A negative value refers to the time before and a positive value refers to the time after.

\$awlaser[CHN]		Description
\$va_on_dly Default: -40 ms		Analog signal ON delay time in msec w/r to a taught point at Laser Start. A negative value refers to the time before and a positive value refers to the time after.
\$va_of_dly Default: 40 ms		Analog signal OFF delay time in msec w/r to a taught point at Laser End. A negative value refers to the time before and a positive value refers to the time after.
\$cmd	\$on_idx	DO index for command output for ON/OFF control.
	\$va_idx	Output I/O index for command output.
	\$type	Output I/O type (Group or Analog).
	\$slop	Output I/O slope.
	\$offs	Output I/O offset.
	\$on_st	Digital Output status for command output.
	\$val	Command output value.
	\$chk_lim Default: 0	Bitwise setup to check different limits: Bit0: control the lower limit. Bit1: control the upper limit. Bit2: control the minimum incremental. Bit3: control the maximum incremental.
	\$min_inc	Specifies minimum incremental control value.
	\$max_inc	Specifies maximum incremental control value.
	\$min_lim	Specifies lower limit relative to command value.
	\$max_lim	Specifies upper limit relative to command value.

\$awlaser[CHN]		Description
\$fbk	\$on_idx	DI index for feedback ON/OFF state. It is specially used by laser channel to check laser on signal during weld. If \$on_idx = 0, system will not check Laser ON signal. If \$on_idx > 0 and DI value is OFF during weld, system will stop weld process and post alarm ARC-404 Lost Laser detect.
	\$va_idx	Input I/O index for feedback.
	\$type	Input I/O type (Group or Analog).
	\$slop	Input I/O slope.
	\$offs	Input I/O offset.
	\$on_st	Input for channel on status.
	\$val	Input for feedback value.
	\$chk_lim	Limit check modes: Bit0: set this bit for monitoring lower limit. Bit1: set this bit for monitoring upper limit.
	\$min_inc	This item specifies minimum incremental value.
	\$max_inc	This item specifies maximum incremental value.
\$gain[sch] Default 1.0	\$min_lim	This item specifies lower limit below the command value.
	\$max_lim	This item specifies upper limit beyond the command value.
		This item specifies adaptive control gain for each schedule.

\$awlaser[CHN]		Description
<i>\$ramp_up[sch]</i> Used in Laser Start	<i>\$mode</i> Default: 3	1: NONE, no ramp up. 2: AUTO, the ramp up time is determined automatically. 3: USER, the ramp up time is specified by user. 4: SPED, the ramp follows the weld speed profile.
	<i>\$time</i>	Specifies ramp time for USER mode.
	<i>\$dly_t</i>	Specifies ramp delay time in msec.
	<i>\$str_v</i>	The user defined beginning value at ramp up
	<i>\$end_v</i>	System determines end value at ramp up.
	<i>\$st_tick (RO)</i>	System records the start tick at beginning of ramp up.
	<i>\$state (RO)</i>	0: not ready. 1: ready to ramp. 2: ramp starts. 3: ramp is done.

\$awlaser[CHN]		Description
<i>\$ramp_dn[sch]</i> Used in Laser End	<i>\$mode</i> Default: 1	1: NONE, no ramp down. 2: AUTO, the ramp time is determined automatically. 3: USER, the ramp time is specified by user. 4: SPED, the ramp follows the weld speed profile.
	<i>\$time</i>	Specifies ramp time for USER mode.
	<i>\$str_v</i>	System determines the beginning value at ramp down.
	<i>\$end_v</i>	The user defined end value of ramp. For wire feed channel, a negative value represents the wire retract speed and enables the wire retract at the end of ramp down.
	<i>\$dly_t</i>	Specifies the wire retract start time in msec w/r to a taught point at Laser End. This parameter will be ignored if <i>\$end_v</i> > 0.
	<i>\$rtk_time</i>	Specifies the wire retract time in msec. This parameter will be ignored if <i>\$end_v</i> > 0.
	<i>\$st_tick (RO)</i>	Start tick at beginning of ramp down.
	<i>\$state (RO)</i>	0: not ready. 1: ramp up. 2: adapt process. 3: ramp dn. 4. ramp schedule 5. wire retract
<i>\$param_f[1-10]</i>		Internal use.
<i>\$param_i[1-10]</i>		Internal use.

11 PROGRAMMING

11.1 PROGRAM EXAMPLE 1 (SINGLE SCHEDULE WITH CORNER MOVES)

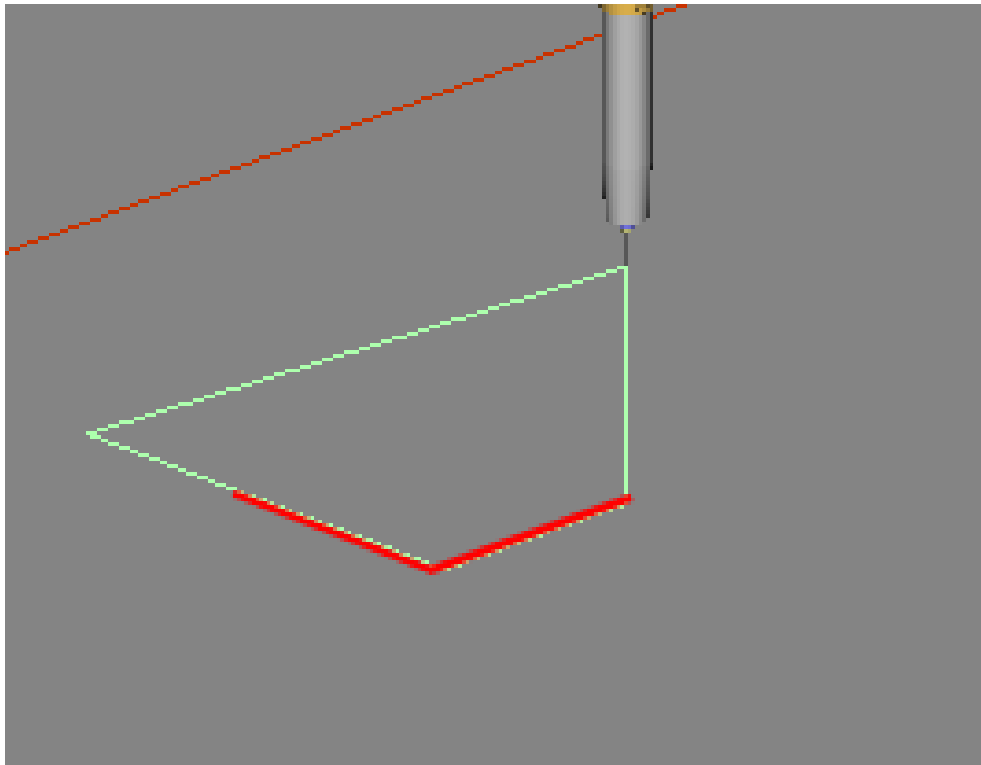


Figure 11.1 (a) Weld Path (linear and corner move)

```
1:L P[1] 800mm/sec FINE;  
2: Sample Start[1];  
3:L P[2] 80.0inch/min FINE  
  : Laser Start[1];  
4:L P[3] WELD_SPEED CNT100;  
5:L P[4] WELD_SPEED FINE  
  : Laser End[1];  
6:L P[5] 20mm/sec FINE;  
7: Sample End;
```

Figure 11.1 (b) Program with one schedule

The chart below illustrates the overall timing (laser and wfs) for the entire program execution.

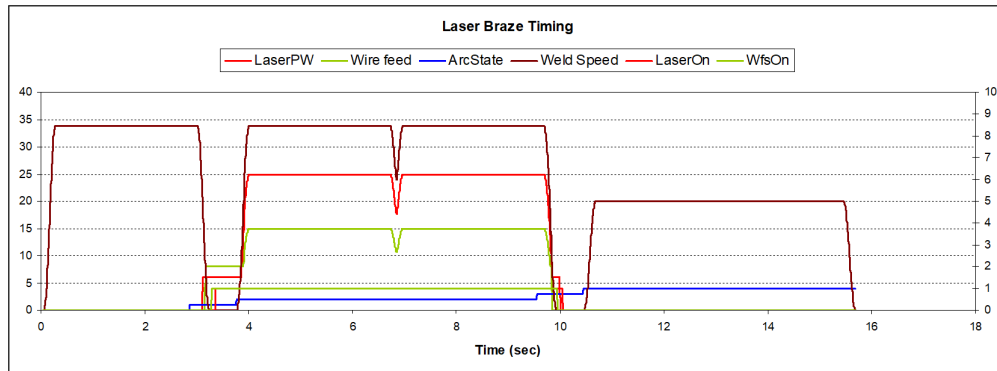


Figure 11.1 (c) Laser Process Control Timing with Standard Wire Feeder

The chart below shows the detail timing at Laser Start.

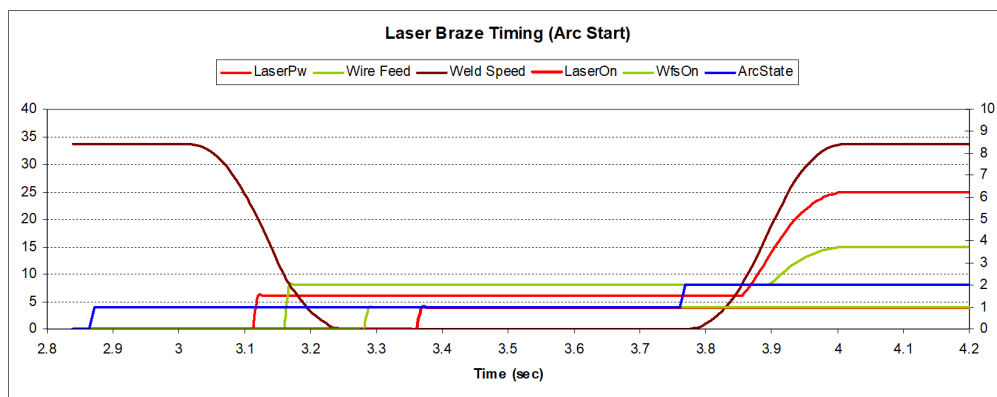


Figure 11.1 (d) Laser Process Control Timing at Laser Start with Standard Wire Feeder

The chart below shows the detail timing at the Laser End.

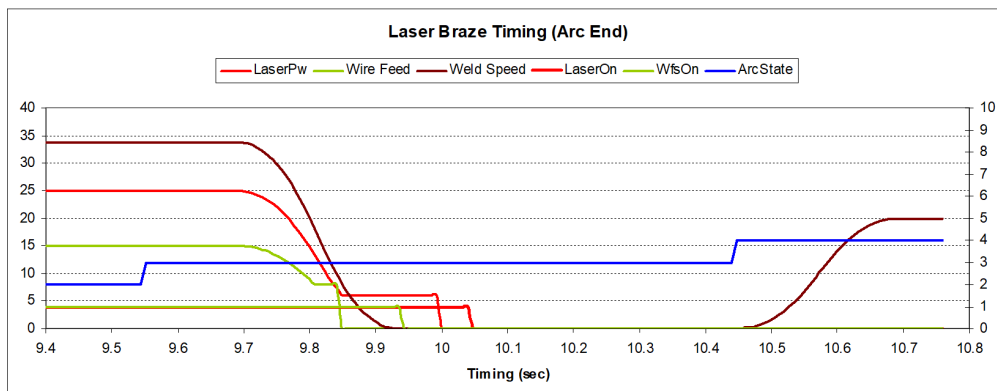


Figure 11.1 (e) Laser Process Control Timing at Laser End with Standard Wire Feeder

11.2 PROGRAM EXAMPLE 2 (SCHEDULE CHANGES)

In this example, the Laser starts with schedule 1 for line 4, changes to schedule 2 for line 6 and schedule 3 for line 8. The weld speed is different for each weld schedule, thus we can use SPED ramp mode for all three schedules.

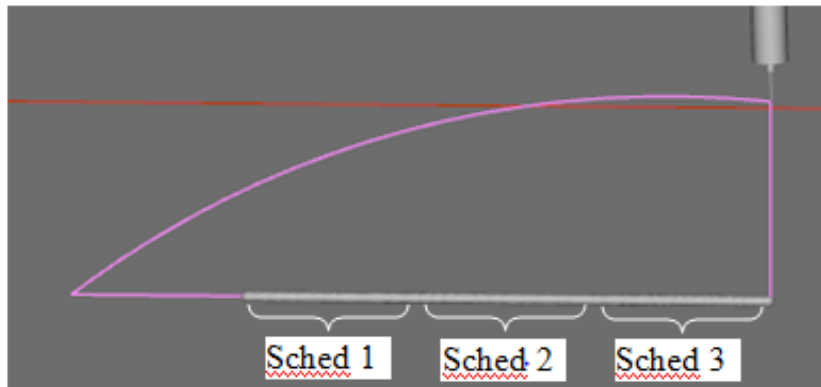


Figure 11.2 (a) Weld Path (schedule changes)

```

1:J P[1] 10% FINE      ;
2:  Sample Start[1] ;
3:L P[2] 100mm/sec FINE
   : Laser Start[1]      ;
4:L P[3] WELD_SPEED CNT100 ;
5:  Laser Start[2] ;
6:L P[6] WELD_SPEED CNT100 ;
7:  Laser Start[3] ;
8:L P[4] WELD_SPEED FINE
   : Laser End[1]      ;
9:L P[5] 100mm/sec FINE;
10:  Sample End ;

```

Figure 11.2 (b) Program with Schedule Changes

DATA Weld Sched				
	WATT	mm/sec	mm/sec	COMMENT
1	40.0	35.0	70	

Figure 11.2 (c) Weld Schedule

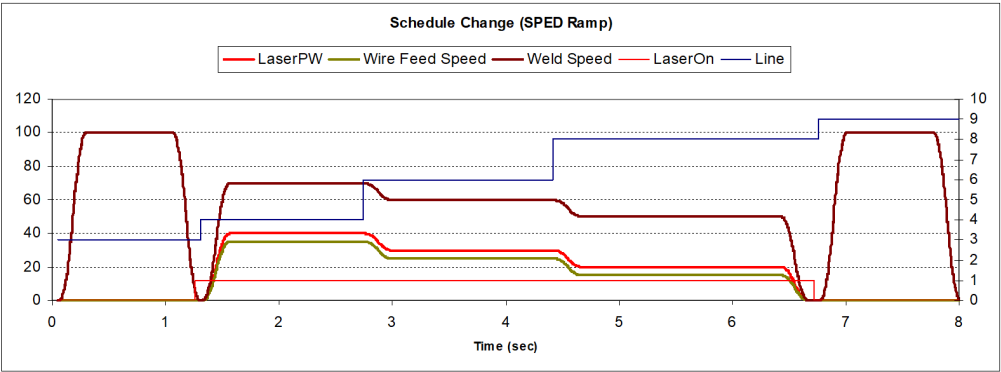


Figure 11.2 (d) Laser Process Control with Schedule Change

12 ERROR CONDITIONS AND ERROR RECOVERY

When faults occur during laser braze/weld operation, the system will turn off the laser and/or wire feed and hold motion. Once the fault condition is cleared, the laser braze/weld operation can be resumed from the fault location. Sometimes, you need to jog the robot from the fault position to a maintenance location for service and the operation can be resumed directly from the service location. The error recovery of laser braze/laser weld will follow the specifications:

- At hold (fault condition), all process signals will be turned off immediately.
- At resume, all signal's timing and ramping control will be the same as the regular Laser Start process.
- At Original Path Resume, the resume operation starts from the service location, the robot will move to the fault position first and then to start the laser weld.

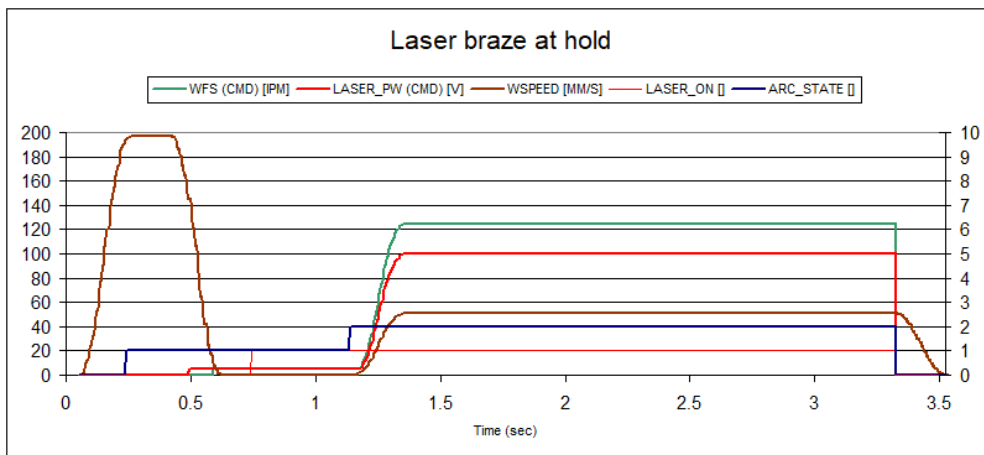


Figure 12 (a) Laser Process Control Timing at Hold

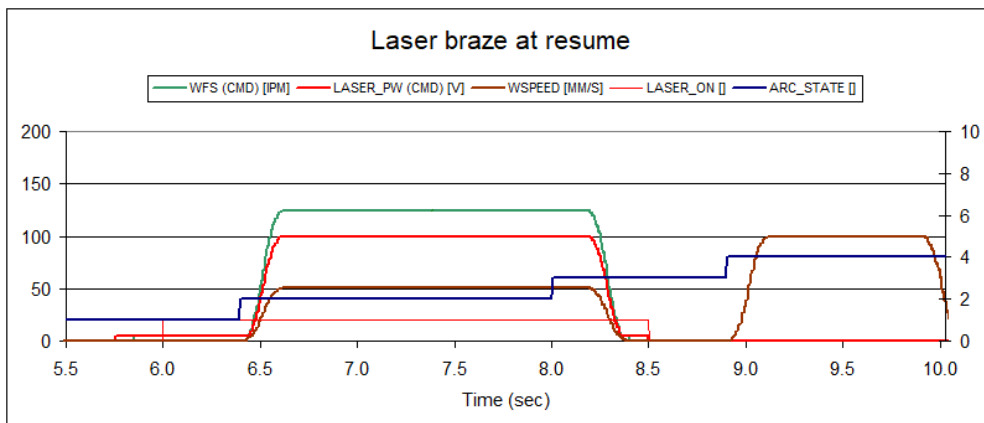


Figure 12 (b) Laser Process Control Timing at Resume

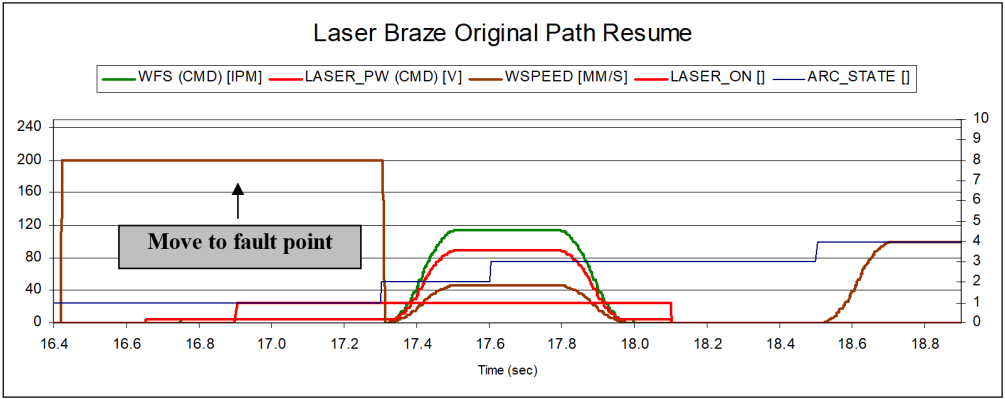


Figure 12 (c) Laser Process Control Timing at OPR

At original Path Resume, the channel control timing and ramp control is the same as the regular Laser Start.

13 ERROR HANDLING

13.1 USER DEFINED ERROR CONDITIONS

Laser Process Control function allows user to trigger CTRL STOP alarm by user defined conditions. This alarm can be triggered by an integer flag value or an digital input.

- **ARC-405** CTRL STOP (Prog, Line)
 - Trigger condition
 1. If $\$awlaser_cfg.\$err_flag1 > 0$ or
 2. If $\$awlaser_cfg.\$st_io_idx > 0$ and $DI[\$awlaser_cfg.\$st_io_idx]$ is OFF
 - Monitor timing: Whenever laser is ON.

13.2 LASER FEEDBACK CHECK

- **ARC -400** Laser PW Up Limit (Prog, Line),
 - Trigger condition: if $\$awlaser[1].\$fbk\$val > \text{Upper limit}$
 - Monitor timing: during adaptive laser welding (after ramp up and before ramp down)
 - System will set bit1 of $\$awlaser[1].\$param_i[1]$ when laser power exceeds the upper limit
 - System will clear bit1 of $\$awlaser[1].\$param_i[1]$ when laser power is below the upper limit
- **ARC -401** Laser PW Lo Limit (Prog, Line)
 - Trigger condition: if $\$awlaser[1].\$fbk\$val < \text{Lower limit}$
 - Monitor timing: during adaptive laser welding (after ramp up and before ramp down)
 - System will set bit0 of $\$awlaser[1].\$param_i[1]$ when laser power is below the lower limit
 - System clear bit0 of $\$awlaser[1].\$param_i[1]$ when laser power is above the lower limit
- **ARC -404** Lost Laser Detect (Prog, Line)
 - Trigger condition: if $DIN[\$awlaser[1].\$fbk\$on_idx] = \text{OFF}$
 - Monitor timing: during laser weld before laser is off.
- **ARC -408** No Laser detected (Prog, Line)
 - Trigger condition: if $DIN[\$awlaser[1].\$fbk\$on_idx] = \text{OFF}$
 - Monitor timing: $\$awlaser[1].\$do_on_dly + \$awlas_cfg.\$wait_time$

13.3 WIRE FEED FEEDBACK CHECK

- **ARC -402** WF Speed Up Limit (Prog, Line)
 - Trigger condition: if $\$awlaser[2].\$fbk\$val > \text{Upper limit}$
 - Monitor timing: during adaptive laser welding (after ramp up, before ramp down)
 - System will set bit1 of $\$awlaser[2].\$param_i[1]$ when wire feed speed exceeds the upper limit
 - System will clear bit1 of $\$awlaser[2].\$param_i[1]$ when wire feed speed is below the upper limit

- **ARC -403** WF Speed Lo Limit (Prog, Line)
- Trigger condition: if $\$awlaser[2].\$fbk\$val < \text{Lower limit}$
- Monitor timing: during adaptive laser welding (after ramp up, before ramp down)
- System will set bit0 of $\$awlaser[2].\$param_i[1]$ when wire feed speed is below the lower limit
- System will clear bit0 of $\$awlaser[2].\$param_i[1]$ when wire feed speed is above the lower limit

14 ALARMS

Table 14 Alarms

Alarm	Severity	Description
ARC-400 Laser PW Up limit (Prog, Line)	Warning	The laser power reaches upper limit
ARC-401 Laser PW Lo limit (Prog, Line)	Warning	The laser power reaches lower limit
ARC-402 WF Speed Up limit (Prog, Line)	Warning	The wire feed speed reaches upper limit
ARC-403 WF Speed Lo limit (Prog, Line)	Warning	The wire feed speed reaches lower limit
ARC-404 Lost Laser detect (Prog, Line)	STOP	Lost laser on signal during welding
ARC-405 CTRL STOP (Prog, Line)	STOP	User defined control stop during welding
ARC-406 Exceeds Max Laser Sch (Prog, Line)	STOP	Exceeds maximum schedule number
ARC-407 No Direct Value Input (Prog, Line)	STOP	No direct value on Laser Start
ARC-408 No Laser detect (Prog, Line)	STOP	No laser detected on at laser welding start

15 I/O CONNECTIONS

15.1 MOCK ARC WELD I/O

Laser Process Control does not use Arc Weld I/O. However, because the Arc Weld I/O signals are checked to exist on power up, it is necessary to assign them to memory image I/O ports (ports that have no actual I/O connection). If any of the Arc Weld I/O signals are not setup correctly, an alarm will be posted. For example: **ARC-040** E1 Missing I/O Weld Start.

Note

The Cause and Remedy for the **ARC-040** alarm suggests using the CONFIG key in the Arc Weld I/O screen to complete the I/O assignment. However, this cannot be done when the system is configured for Laser Process Control.

Memory image I/O ports are memory locations in the robot controller that are not connected to actual I/O ports. The controller has 2500 Boolean (ON/OFF) memory image ports for use as digital I/O and 250 numeric image ports for use as analog or group I/O. Memory image I/O ports are configured in the CONFIG level of the IO screen by specifying rack 0 and slot 0.

Table 15.1 (a) Mock Arc Input Signals

\$AWEPRR[1]	Value	Example
<i>\$voltage_fbk.\$port_num</i>	Mock AI port number	1
<i>\$voltage_fbk.\$port_type</i>	3	
<i>\$current_fbk.\$port_num</i>	Mock AI port number	2
<i>\$current_fbk.\$port_type</i>	3	
<i>\$arc_detect.\$port_num</i>	Mock DI port number	1217
<i>\$arc_detect.\$port_type</i>	1	
<i>\$gas_fault.\$port_num</i>	Mock DI port number	1218
<i>\$gas_fault.\$port_type</i>	1	
<i>\$wire_fault.\$port_num</i>	Mock DI port number	1219
<i>\$wire_fault.\$port_type</i>	1	
<i>\$water_fault.\$port_num</i>	Mock DI port number	1220
<i>\$water_fault.\$port_type</i>	1	
<i>\$pwr_fault.\$port_num</i>	Mock DI port number	1221
<i>\$pwr_fault.\$port_type</i>	1	
<i>\$wstk_input.\$port_num</i>	Mock DI port number	1223
<i>\$wstk_input.\$port_type</i>	1	

Table 15.1 (b) Mock Arc Output Signals

\$AWEPRR[1]	Value	Example
<i>\$voltage_cmd.\$port_num</i>	Mock AO port number	1
<i>\$voltage_cmd.\$port_type</i>	4	
<i>\$wfs_cmd.\$port_num</i>	Mock AO port number	2
<i>\$wfs_cmd.\$port_type</i>	4	
<i>\$weld_start.\$port_num</i>	Mock DO port number	1217
<i>\$weld_start.\$port_type</i>	2	
<i>\$gas_start.\$port_num</i>	Mock DO port number	1218
<i>\$gas_start.\$port_type</i>	2	
<i>\$inch_fwd.\$port_num</i>	Mock DO port number	1219
<i>\$inch_fwd.\$port_type</i>	2	
<i>\$inch_bwd.\$port_num</i>	Mock DO port number	1220
<i>\$inch_bwd.\$port_type</i>	2	
<i>\$wstk_alarm.\$port_num</i>	Mock DO port number	1224
<i>\$wstk_alarm.\$port_type</i>	2	

! WARNING

These \$AWEPRR system variable assignments are re-initialized if the weld process is reconfigured in the ArcTool CTRL-START setup menu.

15.1.1 Configuring the Memory Image IO and Assigning the Arc Weld IO

About this task

Follow this procedure to configure the memory image I/O and assign the Arc Weld I/O.

Procedure

1. Configure the memory image I/O for the analog inputs and outputs.

Use channel numbers that are not in use.

I/O Analog Out				I/O Analog In			
AO #	RACK	SLOT	CHANNEL	AI #	RACK	SLOT	CHANNEL
1	0	0	1	1	0	0	3
2	0	0	2	2	0	0	4

2. Configure the memory image I/O for the digital inputs and outputs.

Use a range of ports that are not in use.

I/O Digital Out					I/O Digital In				
#	RANGE	RACK	SLOT	START STAT.	#	RANGE	RACK	SLOT	START STAT.
12	DO[1193-1216]	0	0	0 UNASG	12	DI[1217-1224]	0	0	9 PEND
13	DO[1217-1224]	0	0	1 PEND	13	DI[1225-2000]	0	0	0 UNASG

3. Re-power the controller.
4. Verify that the digital and analog I/O are now assigned.
5. Assign the Arc Weld I/O by setting the $\$AWEPRR[I]$ system variable fields according to table [Table 15.1 \(a\)](#) and table [Table 15.1 \(b\)](#). Use the analog channel numbers and digital port numbers setup in steps 1 and 2.
6. Re-power the controller.

15.2 I/O CONNECTION TO IPG

Refer to the section of Ethernet/IP interface FieldBus MPI Protocol in high Power Fiber Laser User Guide. The table below lists the I/O connection for commanded robot output to the IPG input via an Ethernet interface.

Table 15.2 (a) Robot output to IPG

Robot Output for IPG		IPG Input		
Output Function	Output assignment	Bit	Byte	Input description
Laser ON (cmd)	DO[\$awlaser[1].\$cmd.\$on_idx]	1	0	This bit is used to switch ON and OFF the Laser (main power supply).
Laser power (cmd)	GO[\$awlaser[1].\$cmd.\$va_idx]	0-15	2	Power of the laser in watts (up to 65kW). 1 Bit = 1 Watt. Decimal coded.
			3	

Refer to the section of Ethernet/IP interface FieldBus MPI Protocol in high Power Fiber Laser User Guide. The table below lists the I/O connection for IPG output to Robot input via an Ethernet interface.

Table 15.2 (b) Robot input from IPG

Robot Input from IPG		IPG Output		
Input Function	Input assignment	Bit	Byte	Output description
Laser ON (fbk)	DI[\$awlaser[1].\$fbk.\$on_idx]	1	0	This bit is used to switch ON and OFF the Laser (main power supply).

Robot Input from IPG		IPG Output		
Laser power (fbk)	GI[\$awlaser[1].\$fbk .\$va_idx]	0-15	2	Measured (real) output power of the laser in watts. Decimal coded.
			3	

15.3 I/O CONNECTION TO LASERLINE

Refer to the input signals section of the *laserline User Guide table 145 and Fig. 213 Activate PRG start, static*. The table below lists the I/O connection for commanded robot output to the laserline input via DeviceNet interface.

Table 15.3 (a) Robot output to laserline

Robot Output for laserline		laserline Input		
Output Function	Output assignment	Bit	Word	Input description
Laser ON (cmd)	DO[\$awlaser[1].\$cmd.\$on_idx]	11	0	Static PRG start. If the signal edge changes from an inactive to an active signal, the current program is started. The program is canceled as soon as the signal switches to an inactive state (if the program has not already been ended).
Laser power (cmd)	GO[\$awlaser[1].\$cmd.\$va_idx]	0-15	3	Words 3 can be used in Program mode for analog value To preset laser power.

Refer to the output signals section in the *laserline User Guide Tab 170, 184 and 185*. The table below lists the I/O connection for laser line output to robot input via DeviceNet interface.

Table 15.3 (b) Robot input from laserline

Robot Input from laserline		laserline Output		
Input Function	Input assignment	Bit	Word	Output description

Robot Input from laserline		laserline Output		
Laser ON (fbk)	DI[\$awlaser[1].\$fbk .\$on_idx]	12	0	PRG active status bit. This signal is set as soon as a laser program is running. It is reset when the program is ended or canceled.
Laser power (fbk)	GI[\$awlaser[1].\$fbk .\$va_idx]	0-15	6	Current laser power as 16-bit integer value (binary coded).

15.4 I/O CONNECTION TO ABICOR BINZEL WIRE FEEDER

Table 15.4 (a) Robot output to Binzel Wire Feeder

Robot Output for Binzel Wire Feeder		Binzel E-Box Input		
Output Function	Output assignment	Bit	Byte	E-Box input
ON/OFF control	DO[\$awlaser[2].\$cmd.\$on_idx]	0	0	Start wire feed
Wire feed backward	DO[\$awlaser_cfg.\$fbw_do_idx]	1	0	Wire feed reverse ¹
		2	0	Initialize ¹
Inching forward	DO[\$awlaser_cfg.\$fw_do_idx]	3	0	Inching forward
Inching backward	DO[\$awlaser_cfg.\$bw_do_idx]	4	0	Inching backward
Wire feed speed	GO[\$awlaser[2].\$cmd.\$va_idx]	0-7	2	Wire feed speed, resolution 10 bits ²
		8-9	3	

Table 15.4 (b) Robot input from Binzel Wire Feeder

Robot input from Binzel Wire Feeder		Binzel E-Box Output		
Input Function	Input assignment	Bit	Byte	E-Box output
Wire feed speed	GI[\$awlaser[2].\$fbk .\$va_idx]	0-7	4	Real wire feed speed, resolution 10 bits ²
		8-9	5	

¹ The wire feed reverse bit is not used and the initialize bit is handled by user program.

² The Group output for wire feed speed is 10bits, and its maximum value is 1023.

INDEX

Special Characters

\$AWLASER[CFG] [35](#)
\$AWLASER[CHN] [36](#)

A

ALARMS [49](#)
ARCTOOL SETUP [1](#)
ARCTOOL SETUP MENU AT CTRL START [9](#)

C

Configuring the Memory Image IO and Assigning the Arc Weld IO [52](#)
CONTROL LOOP TIME [31](#)
CONTROL SIGNAL CHANNELS SPECIFICATION [3](#)

E

Error Conditions and Error Recovery [45](#)
Error Handling [47](#)

F

FILTERING COMMAND CONTROLS [32](#)
FUNCTION OVERVIEW [1](#)

I

I/O CONNECTION TO ABICOR BINZEL WIRE FEEDER [55](#)
I/O CONNECTION TO IPG [53](#)
I/O CONNECTION TO LASERLINE [54](#)
I/O Connections [51](#)
I/O WELD FOR LASER [15](#)

L

Laser Feedback Check [47](#)
LASER PROCESS CONTROL FUNCTION (R828) [1](#)
LASER PROCESS CONTROL SOFTWARE VERSION AND PLATFORM [1](#)
LASER START AND LASER END INSTRUCTIONS [16](#)
LASER START AND LASER END TIMING SPECIFICATION [7](#)
Laser Timing Description [12](#)
LASER TIMING SETUP [11](#)
Laser Timing Setup Menu [11](#)
LASER WELD SCHEDULE DATA SCREEN [9](#)
LASER WELDER AND WIRE FEEDER SPECIFICATION [5](#)
Laser Welding Start Timing [14](#)
LIMITATIONS [2](#)
LINEAR PROPORTIONAL CONTROL ALGORITHM [31](#)

M

MOCK ARC WELD I/O [51](#)
MONITORING FEEDBACK VALUE [33](#)

P

PROCESS ADAPTIVE CONTROL [31](#)
PROGRAM EXAMPLE 1 (SINGLE SCHEDULE WITH CORNER MOVES) [41](#)
Program Example 2 (schedule changes) [42](#)
PROGRAMMING [41](#)

R

RAMP CONTROL AT SCHEDULE CHANGE (\$AWLASER[CHN].\$RAMP_UP[SCH]) [20](#)
RAMP DOWN CONTROL AT LASER END (\$AWLASER[CHN].\$RAMP_DN[SCH]) [19](#)
RAMP DOWN EXAMPLES [21](#)
RAMP SCHEDULE CHANGE EXAMPLE [27](#)
RAMP UP CONTROL AT LASER START (\$AWLASER[CHN]. \$RAMP_UP[SCH]) [19](#)
RAMP UP EXAMPLES [20](#)
RAMPING CONTROL [19](#)
replace with title name [13](#)

S

SYSTEM VARIABLES [35](#)

U

UIF [9](#)
User Defined Error Conditions [47](#)

W

WELD STATUS [16](#)
WIRE FEED FEEDBACK CHECK [47](#)
WIRE RETRACT [25](#)
WIRE RETRACT EXAMPLE [26](#)
WIRE RETRACT SETUP [25](#)
WIRE RETRACT TIMING [25](#)

REVISION RECORD

Edition	Date	Contents
B	Mar., 2021	•

